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## Methods

# Matching Impatient and Heterogeneous Demand and Supply

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**Abstract.** Service platforms must determine rules for matching heterogeneous demand (customers) and supply (workers) that arrive randomly over time and may be lost if forced to wait too long for a match. Our objective is to maximize the cumulative value of matches, minus costs incurred when demand and supply wait. We develop a fluid model, that approximates the evolution of the stochastic model and captures explicitly the nonlinear dependence between the amount of demand and supply waiting and the distribution of their patience times, also known as reneging or abandonment times in the literature. The fluid model-invariant states approximate the steady-state mean queue lengths in the stochastic system and, therefore, can be used to develop an optimization problem whose optimal solution provides matching rates between demand and supply types that are asymptotically optimal (on fluid scale as demand and supply rates grow large). We propose a discrete review matching policy that asymptotically achieves the optimal matching rates. We further show that, when the aforementioned matching optimization problem has an optimal extreme point solution, which occurs when the patience time distributions have increasing hazard rate functions, a state-independent priority policy that ranks the edges on the bipartite graph connecting demand and supply is asymptotically optimal. A key insight from this analysis is that the ranking critically depends on the patience time distributions and may be different for different distributions even if they have the same mean, demonstrating that models assuming, for example, exponential patience times for tractability, may lack robustness. Finally, we observe that, when holding costs are zero, a discrete review policy that does not require knowledge of interarrival and patience time distributions is asymptotically optimal.

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**Supplemental Material:** The online appendix is available at <https://doi.org/10.1287/opre.2022.0005>.

**Keywords:** matching • two-sided platforms • bipartite graph • discrete review policy • high-volume setting • fluid model • impatience • reneging • abandonment

## 1. Introduction

Service platforms exist to match demand (customers) and supply (workers); see, for example, Hu (2019) for a broader perspective on the intermediary role of platforms in the sharing economy. The challenge in operating these platforms is that demand and supply often come from heterogeneous customers and workers who arrive randomly over time and do not like to wait. Then, there is a trade-off between making valuable matches quickly and waiting in order to enable better matches. Our goal is to develop matching policies that optimize a long-run objective function consisting of the

total value of matches made minus holding costs incurred when demand and supply wait.

The dislike of waiting can lead to demand and supply abandoning before being matched, for example, as occurs in a ride-hailing application (Wang et al. 2023) in which customers must be matched to workers (drivers). Moreover, the willingness of agents<sup>1</sup> to wait can change over time with some agents becoming more patient (for example, as a result of having already paid a waiting cost) and some becoming less patient (for example, as a result of becoming more and more frustrated when waiting). In a transplant application

setting, in which donors must be matched with needy patients, the survival time of a patient depends on the history and not only on the current state of health (Bertsimas et al. 2019). Also, in a call center setting, empirical evidence does not usually support using an exponential distribution to fit customer patience times (Brown et al. 2005, Mandelbaum and Zeltyn 2013). As a result, using an exponential distribution to model agent patience times may be too restrictive. Still, the exponential distribution is more analytically tractable, and if optimal or near-optimal matching policies are robust to distributional characteristics, then the exponential distribution is a good modeling choice.

The issue when testing whether the aforementioned robustness holds is that solving for an optimal matching policy appears intractable for nonexponential patience time distributions. This is because, without the memoryless property, in order for the system to be Markovian, the state space must track the amount of time each agent has waited, resulting in a very complex state space. To overcome this, we study a deterministic fluid model, that serves as an approximation to the stochastic model when arrival rates are large. The invariant states of the fluid model (that is, the fixed points of the fluid model equations) approximate the steady-state mean amount of demand and supply waiting to be matched and depend nonlinearly on the patience time distributions. We use the aforementioned invariant states to define an optimization problem, termed the matching problem (MP), whose objective aims to maximize the value of matches made minus holding costs. Then, we study the effect of the patience time distribution on the MP solution.

Figure 1 shows that the choice of distribution used to model agent patience times can change how the MP solution prioritizes matches. For the simple matching network in Figure 1(a), the MP solution either

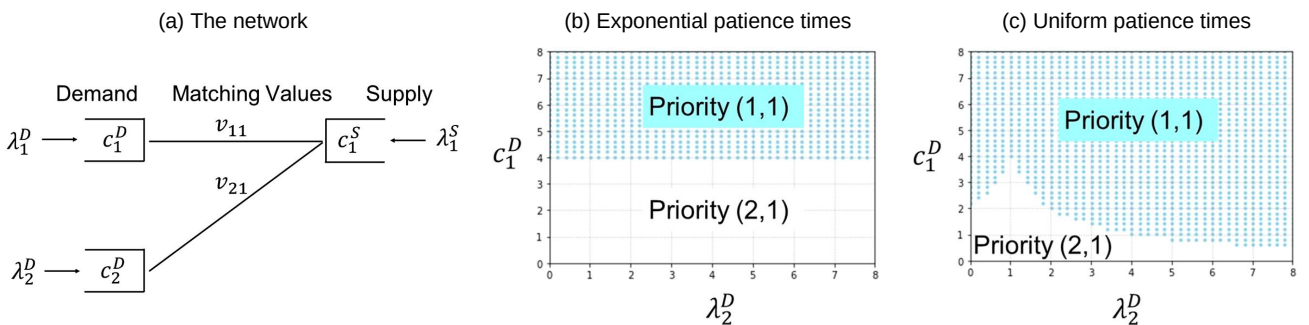
prioritizes making matches on edge (1, 1) before using any leftover supply for making matches on the edge (2,1) or vice versa. Figure 1, (b) and (c), shows that the same parameters lead to different priority orderings for the exponential and uniform patience time distributions. As a result, we are motivated to develop methodology that allows us to analyze as general a class of patience time distributions as possible; in particular, the assumptions we require to justify the MP are that the patience time distributions have a density, a finite mean, and a strictly increasing cumulative distribution function (cdf).<sup>2</sup>

The matching model we consider generalizes that shown in Figure 1 to allow for an arbitrary number of demand and supply types, each with a possibly different patience time distribution and arrival rate. The matching values between demand and supply types are the edge values on the underlying bipartite graph, which can be positive or zero. Because different edges have different values, some matches are preferable to others. Because different demand and supply types have different holding costs and patience time distributions, waiting is more costly for some types than others. Our objective captures the trade-off between these opposing priorities by incorporating both matching values and holding costs.

### 1.1. Contributions of This Paper

We contribute a number of theoretical results and practical insights for matching on service platforms. First, the MP provides an asymptotic upper bound on the objective function value as arrival rates become large (Theorem 3). We then develop a discrete review matching policy whose matching rates mimic an MP optimal solution (Theorem 4) and is asymptotically optimal as arrival rates become large. As a consequence, because an MP optimal solution depends on the patience time

**Figure 1.** (Color online) The Priority Matches Given by the MP Solution for a Network with One Supply Node and Two Demand Nodes



*Note.* The nodes have matching values  $v_{11} = v_{21} = 1$  (i.e., matching a supply unit with a demand unit from either node gives a value of one), identical patience time distributions for all types having mean equal to one, holding costs per job per unit time  $c_2^D = 4$ ,  $c_1^S = 1$ , and Poisson arrivals with rates per unit time  $\lambda_1^D = \lambda_1^S = 1$ .

distributions, so does the proposed matching policy. Whereas this policy is applicable for general patience time distributions, our analysis provides further insight for the important special case when there exists an optimal extreme point solution to the MP (this is ensured when the patience time hazard rates are increasing, i.e., supply and demand become more impatient the longer they wait), which we discuss next.

When the patience time hazard rate functions are increasing, the MP becomes a convex maximization problem and so has an optimal extreme point solution (Theorem 1). In this case, we propose a discrete review policy that is based on a static ranking of the edges and makes matches by prioritizing the use of demand and supply according to that ranking. We show that this policy also achieves the matching rates given by the MP (Theorem 5) and is asymptotically optimal as arrival rates become large. The static ranking depends on the patience time distribution and can be different for different distributions with the same mean as observed in Figure 1. As a result, the service platform must obtain distributional information to determine edge priorities; knowing only information about the mean is not enough.

In the case that holding costs are zero, we observe that an MP optimal solution does not depend on the patience time distribution. Then, a discrete review policy based on a linear problem (LP) that does not need to know information about the arrival rates or the patience time distribution is asymptotically optimal as arrival rates become large (Theorem 6).

The MP is based on the invariant states of a fluid model (Definition 2) as mentioned earlier. Our asymptotic optimality results mentioned in the previous paragraphs require showing that the fluid model starting from any initial state converges to an invariant state as time becomes large (Theorem 2). Our convergence proof is nontrivial when viewed in light of the fact that such a result for the fluid model approximating the single-class, many-server queue with impatience was only shown recently in Atar et al. (2023) (despite being worked on for many years) because dealing with the measure tracking the age in service is complicated. In our analysis, we observe that the aforementioned difficulty is not present in matching systems because, when service is instantaneous, the need for an age-in-service measure is eliminated (an observation also leveraged in Ding et al. 2021).

The remainder of the paper is organized as follows. We end Section 1 by reviewing related literature. In Section 2, we provide a detailed model description. In Section 3, we propose our matching policies. A fluid model is presented in Section 4. In particular, we present asymptotic approximations for the stationary mean queue lengths, and we show the impact of the patience time distributions. In Section 5, we introduce our

high-volume setting, and we study the asymptotic behavior of our proposed matching policies. In Section 6, we demonstrate numerically that our policies perform well in both high-volume and nonasymptotic simulations. In Section 7, we study a platform without holding costs, and we propose a matching policy that does not require the knowledge of the system parameters. Concluding remarks are made in Section 8. All the proofs are given in the online appendix.

## 1.2. Literature Review

We focus on on-demand service platforms that aim to facilitate matching. We refer the reader to Benjaafar and Hu (2020), Chen et al. (2020), and Hu (2021) for excellent higher level perspectives on how such platforms fit into the sharing economy and to Hu (2019) for a survey of recent sharing-economy research in operations management. Three important research questions for such platforms identified in Benjaafar and Hu (2020) are how to price services, pay workers, and match requests. These decisions are ideally made jointly; however, because the joint problem is difficult, the questions are often attacked separately. In this paper, we focus on the matching question.

Our basic matching model is a bipartite graph with demand on one side and supply on the other side. There is a long history of studying two-sided matching problems described by bipartite graphs, beginning with the stable matching problem introduced in the groundbreaking work of Gale and Shapley (1962) and continuing to this day; see Abdulkadiroglu and Sönmez (2013) and Roth and Sotomayor (1990) for later surveys and Ashlagi et al. (2020) for more recent work. In the aforementioned literature, much attention is paid to eliciting agent preferences because the outcomes of the matching decisions, made at one prearranged point in time, can be life-changing events (for example, the matches between medical schools and potential residents). In contrast, many platform matching applications are not life-changing events, so there is less need to focus on eliciting agent preference. Moreover, supply and demand often arrive randomly and continuously over time, must be matched dynamically over time as the arrivals occur (as opposed to at one prearranged point in time), and leave the system without being matched (renege) if made to wait too long.

Recent work uses dynamic two-sided matching models with renege in the context of organ allocation (Zenios 1999, Boxma et al. 2011, Ding et al. 2021, Khademi and Liu 2021), trading systems known as crossing networks (Afèche et al. 2014), online dating and labor markets (Arnosti et al. 2021, Kanoria and Saban 2021), ridesharing (Banerjee et al. 2018, Özkan 2020, Özkan and Ward 2020), and quantum switches (Zubeldia et al. 2022). Simultaneously and motivated by the aforementioned applications, there are many works that begin



with a more loosely motivated modeling abstraction, such as Büke and Chen (2017), Liu (2019), Blanchet et al. (2022), Castro et al. (2020), Jonckheere et al. (2023), Masanet and Moyal (2024), and Kohlenberg and Gurvich (2023). The aforementioned works focus on a range of issues; however, except for Boxma et al. (2011), Ding et al. (2021), and Khademi and Liu (2021), all assume that agents’ waiting times are either deterministic or exponentially distributed. Allowing for more general willingness-to-wait distributions is important because, in practice, the time an agent is willing to wait to be matched may depend on how long that agent has already waited. The difficulty is that tracking how the system evolves over time requires tracking the remaining time each agent present in the system will continue to wait to be matched, resulting in an infinite-dimensional state space. In contrast to our work, Boxma et al. (2011) and Khademi and Liu (2021) have one demand and one supply type, and Ding et al. (2021) considers an index policy class.

Taking an approach similar in spirit to how stochastic processing networks are controlled in the queueing literature developed in Harrison (1996, 2006) and Maglaras (1999, 2000), we use a high-volume asymptotic regime to prove that our matching policies are asymptotically optimal. In particular, we begin by solving a static matching problem, and we focus on fluid-scale asymptotic optimality results. Adopting an approach reminiscent of the one Gurvich and Ward (2014) use in a dynamic multipartite matching model that assumes agents will wait forever to be matched, we use a discrete review policy to balance the trade-off between having agents wait long enough to build up matching flexibility but not so long that many will leave.

We consider an objective function that involves the queue lengths and is an infinite horizon objective. As a

result, we need to consider the steady-state performance of the system. This is the focus of the infinite bipartite matching queueing models considered, for example, in Adan and Weiss (2012, 2014), Adan et al. (2018), Afeche et al. (2022), Bušić et al. (2013), Caldentey et al. (2009), Fazel-Zarandi and Kaplan (2018), and Diamant and Baron (2019).

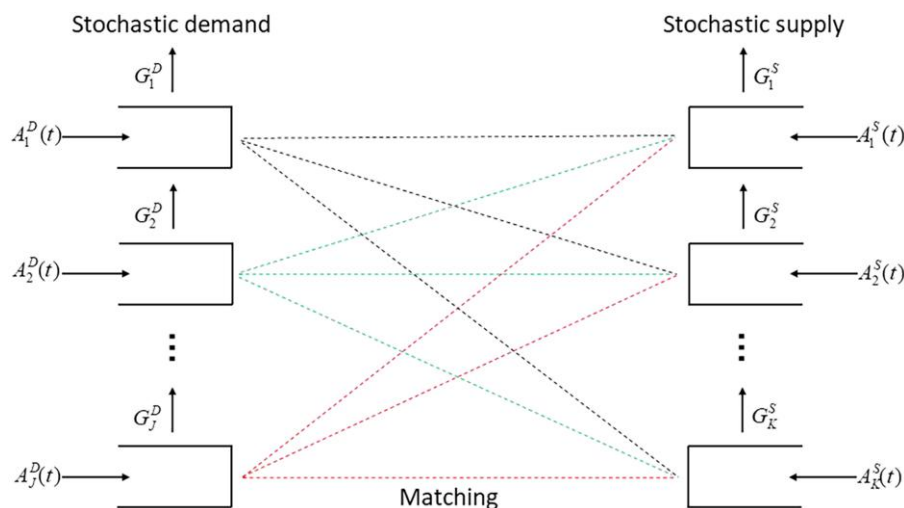
## 2. Model Description

We use the following notation conventions. All vectors and matrices are denoted by bold letters. Further,  $\mathbb{R}$  is the set of real numbers,  $\mathbb{R}_+$  is the set of nonnegative real numbers,  $\mathbb{N}$  is the set of strictly positive integers, and  $\mathbb{Z}_+ = \mathbb{N} \cup \{0\}$ .

## 2.1. Primitive Inputs

The set of demand nodes  $\mathbb{J} := \{1, \dots, J\}$  (representing customer types) and supply nodes (representing worker types)  $\mathbb{K} := \{1, \dots, K\}$  form a bipartite graph as shown in Figure 2. The set  $\mathcal{E} \subseteq \mathbb{J} \times \mathbb{K}$  denotes the set of compatible matches between demand and supply nodes; that is, demand type  $j \in \mathbb{J}$  can be matched with supply type  $k \in \mathbb{K}$  if and only if  $(j, k) \in \mathcal{E}$ . Further, for demand type  $j \in \mathbb{J}$ , let  $\mathcal{S}_j = \{k \mid (j, k) \in \mathcal{E}\}$  denote the set of supply types compatible with  $j$ , and similarly, for supply type  $k \in \mathbb{K}$ , let  $\mathcal{D}_k = \{j \mid (j, k) \in \mathcal{E}\}$  denote the set of demand types compatible with  $k$ . The value of matching demand type  $j \in \mathbb{J}$  and supply type  $k \in \mathbb{K}$  is  $v_{jk} \geq 0$ . The holding costs  $c_j^D \geq 0$  and  $c_k^S \geq 0$  are incurred for each unit of time demand type  $j \in \mathbb{J}$  or supply type  $k \in \mathbb{K}$  waits. Agents arrive according to renewal processes denoted by  $A_j^D(\cdot)$ ,  $j \in \mathbb{J}$  and  $A_k^S(\cdot)$ ,  $k \in \mathbb{K}$ , having respective rates  $\lambda_j^D > 0$  and  $\lambda_k^S > 0$  and interarrival distributions with a finite fifth moment (an assumption convenient for our asymptotic analysis but not strictly required). Upon arrival, each type  $j \in \mathbb{J}$  customer and type  $k \in \mathbb{K}$  worker

**Figure 2.** The Two-Sided Matching Model with General Patience Time Distributions



independently samples patience time from respective distributions  $G_j^D(\cdot)$  and  $G_k^S(\cdot)$  having support on  $[0, H_j^D)$  and  $[0, H_k^S)$  for some  $H_j^D \in [0, \infty]$  and  $H_k^S \in [0, \infty]$ , which represents the maximum amount of time they are willing to wait to be matched. Agents who wait longer than their patience time leave the system without being matched. The patience times are absolutely continuous random variables with density functions  $g_j^D(\cdot)$  and  $g_k^S(\cdot)$  that are mutually independent of each other and of the arrival processes  $A^D$  and  $A^S$ . We let  $h_j^D(x) = g_j^D(x)/(1 - G_j^D(x))$  for  $x \in [0, H_j^D)$  and  $h_k^S(x) = g_k^S(x)/(1 - G_k^S(x))$  for  $x \in [0, H_k^S)$  be the associated hazard functions.

Throughout the paper, we assume that the patience time distributions satisfy the following conditions:

1.  $\int_0^\infty (1 - G_j^D(x))dx = 1/\theta_j^D \in (0, \infty)$ ,  $j \in \mathbb{J}$ , and  $\int_0^\infty (1 - G_k^S(x))dx = 1/\theta_k^S \in (0, \infty)$ ,  $k \in \mathbb{K}$ .
2.  $G_j^D$ ,  $j \in \mathbb{J}$ , and  $G_k^S$ ,  $k \in \mathbb{K}$  are strictly increasing.

The first condition ensures that the patience time distributions have finite positive mean, and the second ensures that the inverse functions  $(G_j^D)^{-1} : [0, 1) \rightarrow [0, H_j^D)$  and  $(G_k^S)^{-1} : [0, 1) \rightarrow [0, H_k^S)$  are well-defined for each  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ . The excess life distributions of the patience time distributions can be defined under the first and are as follows:

$$G_{e,j}^D(x) = \int_0^x \theta_j^D (1 - G_j^D(u))du, \text{ for } j \in \mathbb{J} \text{ and } x \in \mathbb{R}_+$$

and

$$G_{e,k}^S(x) = \int_0^x \theta_k^S (1 - G_k^S(u))du, \text{ for } k \in \mathbb{K} \text{ and } x \in \mathbb{R}_+.$$

The patience time distributions are also known as reneging distributions in the literature. Then, the cumulative number of agents who leave without being matched is more succinctly termed the cumulative reneging. Throughout this paper, we fluctuate between the terms “patience” and “reneging.”

## 2.2. Objective and Admissible Policies

The objective is to maximize the long-run average value of matches made minus holding costs incurred for waiting. A matching process is an  $|\mathcal{E}|$ -dimensional stochastic process  $M(\cdot) := \{M_{jk}(\cdot), (j, k) \in \mathcal{E}\}$  that specifies the cumulative number of matches made between types  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$  in the time interval  $[0, t]$ , denoted by  $M_{jk}(t)$ , for each  $t \geq 0$ , under the assumption that matches are made first come, first served (FCFS) within each type. Then, if  $Q_j^D(t)$  represents the number of type  $j \in \mathbb{J}$  customers waiting and  $Q_k^S(t)$  represents the number of type  $k \in \mathbb{K}$  workers waiting at time  $t \geq 0$ , we want to maximize

$$V_M := \liminf_{t \rightarrow \infty} \frac{1}{t} V_M(t), \quad (1)$$

where

$$V_M(t)$$

$$:= \sum_{(j,k) \in \mathcal{E}} v_{jk} M_{jk}(t) - \sum_{j \in \mathbb{J}} \int_0^t c_j^D Q_j^D(s)ds - \sum_{k \in \mathbb{K}} \int_0^t c_k^S Q_k^S(s)ds.$$

Objective (1) captures the trade-off between making matches quickly and waiting for better matches. The difficulty when deciding whether to incur holding costs for a short period of time in order to enable a potential future higher value match is that the matching policy does not know how much longer each agent currently waiting will remain waiting without being matched.

## 2.3. System State

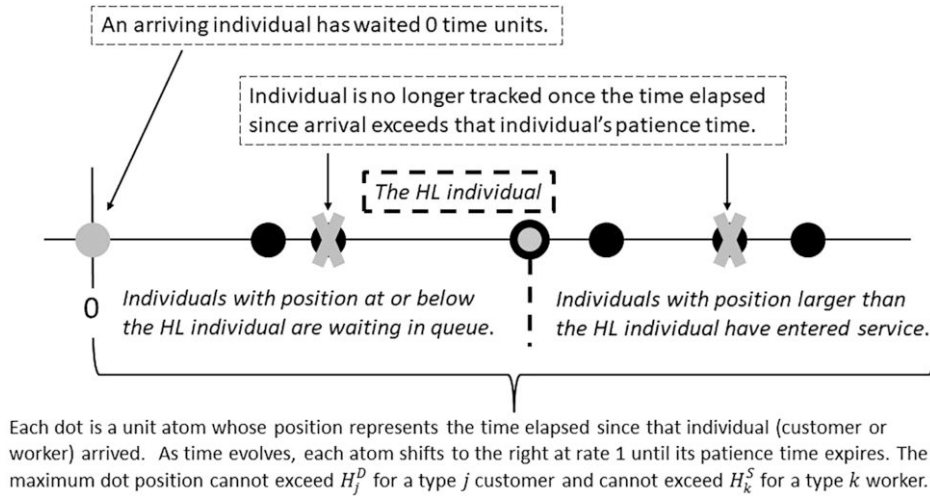
The system state in our model is complex because the state must track how long each agent in the queue has been waiting as well as the time passed since the last arrival of each agent type in order to be Markovian. In particular, tracking how long each agent in the queue has been waiting requires the use of a measure. For  $H \in \mathbb{R}_+ \cup \{\infty\}$ , let  $\mathcal{M}[0, H)$  be the set of finite nonnegative Borel measures on  $[0, H)$ , endowed with the topology of weak convergence. A state at time  $t \in \mathbb{R}_+$  in our model is described as follows: for each  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ ,

- $\alpha_j^D(t) \in \mathbb{R}_+$  and  $\alpha_k^S(t) \in \mathbb{R}_+$  are the times elapsed since the last type  $j$  customer arrived and last type  $k$  worker arrived.
- $Q_j^D(t) \in \mathbb{Z}_+$  and  $Q_k^S(t) \in \mathbb{Z}_+$  are the number of type  $j$  customers and type  $k$  workers waiting in the queue.
- $\eta_j^D(t) \in \mathcal{M}[0, H_j^D)$  and  $\eta_k^S(t) \in \mathcal{M}[0, H_k^S)$  store the amount of time that has passed between each type  $j$  customer’s arrival time and each type  $k$  worker’s arrival time up until that customer’s or worker’s sampled patience time.

The measures  $\eta_j^D$  and  $\eta_k^S$  track the evolution of unit atoms over time, and each atom is associated with a particular type  $j$  customer’s or type  $k$  worker’s time since arrival as shown in Figure 3. All agents tracked are potentially waiting in queue because their wait is less than their sampled patience time. The term “potential” refers to the fact that such agents may or may not have been matched. The FCFS matching assumption implies that all potential customers (workers) who have waited longer than the customer (worker) at the head of the line for their type have already been matched and all potential customers (workers) who have waited less than that customer are in queue. As such, at any time  $t \in \mathbb{R}_+$ , for each  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ ,

$$Q_j^D(t) \leq \int_0^{H_j^D} \eta_j^D(t)dt \text{ and } Q_k^S(t) \leq \int_0^{H_k^S} \eta_k^S(t)dt. \quad (2)$$

The measures  $\eta_j^D(t)$  and  $\eta_k^S(t)$  are independent of the matching process policy, which facilitates the analysis.

**Figure 3.** The Measure-Valued Process  $\eta_j$  (for a Given Customer Type) or  $\eta_k$  (for a Given Worker Type)

## 2.4. Queue Evolution Equations

For a given matching process,  $M(\cdot)$ , the demand and supply queue lengths at time  $t \geq 0$  are given by

$$Q_j^D(t) := Q_j^D(0) + A_j^D(t) - R_j^D(t) - \sum_{k \in S_j} M_{jk}(t) \geq 0 \quad (3)$$

and

$$Q_k^S(t) := Q_k^S(0) + A_k^S(t) - R_k^S(t) - \sum_{j \in D_k} M_{jk}(t) \geq 0, \quad (4)$$

for all  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ , where  $R_j^D(t)$  and  $R_k^S(t)$  denote the cumulative number of type  $j \in \mathbb{J}$  customers and type  $k \in \mathbb{K}$  workers who left the system without being matched in  $[0, t]$ . Equations (3) and (4) are balance equations that say the number of type  $j \in \mathbb{J}$  customers and type  $k \in \mathbb{K}$  workers waiting to be matched at time  $t \geq 0$  equals those present initially plus the cumulative arrivals minus the cumulative reneging minus the cumulative matches.

We refer to  $R^D(\cdot)$  and  $R^S(\cdot)$  as the reneging processes because, in the queueing literature, leaving the system before being served (matched in our context) is commonly known as reneging. We do not explicitly construct these processes, nor do we specify the system dynamics because that involves substantial mathematical overhead that is not relevant to our goal of maximizing (1). Instead, we refer the reader to the companion technical paper Aveklouris et al. (2023) for that. To advance our goal of maximizing (1), we first accept that finding an exact solution does not appear possible and second focus on studying a deterministic matching problem (see (7)) whose objective function approximates (1).

## 2.5. Admissible Matching Policy

A matching policy induces a matching process  $M$ , and that may restrict the set of achievable system states. For

example, if the matching policy prioritizes matches between type  $j \in \mathbb{J}$  customers and type  $k \in \mathbb{K}$  workers, then the matching policy will disallow states in which both type  $j$  customers and type  $k$  workers are present, meaning that  $Q_j^D(t)Q_k^S(t) = 0$  for all  $t \geq 0$ .

A matching policy is admissible if the induced matching process  $M$  satisfies the following natural assumptions. First, no partial matches can be made, and if a match occurs, then it cannot be taken back. That is, a matching process is integer-valued and non-decreasing. Next, matches are recorded exactly when made (not before), meaning a matching process is right-continuous with left limits. Third, agents must be present in the system to be matched, which is equivalent to having nonnegative agent queue lengths as required in (3) and (4). Fourth, a matching process is nonanticipating, meaning the process cannot know the exact times of future arrivals. We call matching processes that satisfy all the aforementioned properties admissible, and the maximization in (1) is over the class of admissible matching processes.<sup>3</sup>

## 3. Proposed Matching Policies

In this section, we develop matching policies with the goal of asymptotically maximizing our objective (1). To do so, we first define an optimization problem that determines the optimal matching rates (Section 3.1), and we refer to it as the MP. Then, we propose a discrete review matching policy that asymptotically achieves these rates (Section 3.2), and we call this policy the matching rate-based policy. When the existence of an optimal extreme point solution to the MP is ensured, we are able to propose a discrete review priority-ordering policy that prioritizes the edges and also asymptotically achieves the optimal matching rates (Section 3.3).

A discrete review policy decides on matches at review time points and does nothing at all other times (see, for example, Gurvich and Ward 2014). Longer review periods allow more flexibility in making matches but risk losing impatient agents. Shorter review periods prevent customer/worker loss and reduce holding cost but may not have sufficient numbers of customers/workers of each type to ensure the most valuable matches can be made.

Let  $t > 0$ . We let  $l, 2l, 3l, \dots$  be the discrete review time points, where  $l \in (0, t)$  is the review period length. Because a discrete review policy does not make matches between review time points, at time  $il$  for  $i \in \{1, \dots, \lfloor t/l \rfloor\}$ , from (3), the number of type  $j$  customers available to be matched is

$$Q_j^D(il-) = Q_j^D((i-1)l) + A_j^D(il) - A_j^D((i-1)l) - R_j^D(il) + R_j^D((i-1)l), \quad (5)$$

for each  $j \in \mathbb{J}$ , and from (4), the number of type  $k$  workers available to be matched is

$$Q_k^S(il-) = Q_k^S((i-1)l) + A_k^S(il) - A_k^S((i-1)l) - R_k^S(il) + R_k^S((i-1)l), \quad (6)$$

for each  $k \in \mathbb{K}$ .

### 3.1. A Matching Problem

Suppose we ignore the discrete and stochastic nature of agent arrivals and assume that these flow at their long-run average rates. Then, an upper bound on the long-run average matching value in (1) follows by solving an optimization problem to find the optimal instantaneous matching values, which we refer to as the MP. For  $\mathbf{m} = (m_{jk} : (j, k) \in \mathcal{E})$ , which denotes the instantaneous matching rate, the MP can be defined through functions  $q_j^{D,*}(\mathbf{m})$ ,  $j \in \mathbb{J}$  and  $q_k^{S,*}(\mathbf{m})$ ,  $k \in \mathbb{K}$  (defined in detail subsequently) that approximate the steady-state mean queue lengths as follows:

$$\begin{aligned} \max \quad & \sum_{(j,k) \in \mathcal{E}} v_{jk} m_{jk} - \sum_{j \in \mathbb{J}} c_j^D q_j^{D,*}(\mathbf{m}) - \sum_{k \in \mathbb{K}} c_k^S q_k^{S,*}(\mathbf{m}) \\ \text{s.t.} \quad & \sum_{j \in \mathcal{D}_k} m_{jk} \leq \lambda_k^S, k \in \mathbb{K}, \\ & \sum_{k \in \mathcal{S}_j} m_{jk} \leq \lambda_j^D, j \in \mathbb{J}, \\ & m_{jk} \geq 0, (j, k) \in \mathcal{E}. \end{aligned} \quad (7)$$

The objective function in (7) is the instantaneous version of the long-run average matching value in (1), and the constraints in (7) prevent us from matching more demand or supply than is available.

An optimal solution to the MP (7),  $\mathbf{m}^* = (m_{jk}^* : (j, k) \in \mathcal{E})$ , can be interpreted as the optimal instantaneous

rate of matches between demand  $j \in \mathbb{J}$  and supply  $k \in \mathbb{K}$ . An admissible matching policy should maximize the long-run average matching value in (1) if its associated matching rates equal  $\mathbf{m}^*$ .

Unfortunately, having closed-form expressions for the exact steady-state mean queue lengths appears intractable. Fortunately, when demand and supply arrival rates become large, we can approximate the steady-state mean queue lengths by developing a fluid model to approximate the evolution of the stochastic model defined in Section 2 and finding its invariant states, which we do in Section 4 (see (25) and (26) in Proposition 3, duplicated as follows for convenience) in order to develop the approximating functions

$$q_j^{D,*}(\mathbf{m}) = \begin{cases} \frac{\lambda_j^D}{\theta_j^D}, & \text{if } \sum_{k \in \mathcal{S}_j} m_{jk} = 0, \\ \frac{\lambda_j^D}{\theta_j^D} G_{e,j}^D \left( (G_j^D)^{-1} \left( 1 - \frac{\sum_{k \in \mathcal{S}_j} m_{jk}}{\lambda_j^D} \right) \right), & \text{if } \sum_{k \in \mathcal{S}_j} m_{jk} \in (0, \lambda_j^D], \end{cases}$$

$$q_k^{S,*}(\mathbf{m}) = \begin{cases} \frac{\lambda_k^S}{\theta_k^S}, & \text{if } \sum_{j \in \mathcal{D}_k} m_{jk} = 0, \\ \frac{\lambda_k^S}{\theta_k^S} G_{e,k}^S \left( (G_k^S)^{-1} \left( 1 - \frac{\sum_{j \in \mathcal{D}_k} m_{jk}}{\lambda_k^S} \right) \right), & \text{if } \sum_{j \in \mathcal{D}_k} m_{jk} \in (0, \lambda_k^S], \end{cases}$$

for each  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ , recalling that  $G_{e,j}^D$  and  $G_{e,k}^S$  are the excess life distributions of the patience times.

An optimal solution to the MP (7) provides a good approximation of the optimal instantaneous matching values if  $q_j^{D,*}(\mathbf{m})$  and  $q_k^{S,*}(\mathbf{m})$  provide good approximations to the steady-state mean queue lengths. Figure 4 provides supporting evidence for this claim in a network with one demand and one supply node that makes every possible match (in which case  $\mathbf{m}$  is one-dimensional and equal to  $\min\{\lambda_1^D, \lambda_1^S\}$ ).

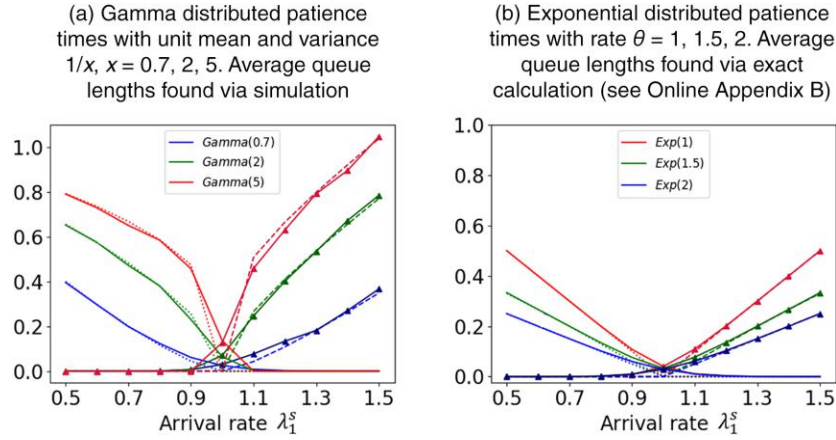
In general, the MP is nonconvex because the approximations of the queue lengths depend on the patience time distributions through a nonlinear relationship. The only time the relationship is linear is when the patience times follow an exponential distribution, namely,

$$q_j^{D,*}(\mathbf{m}) = \frac{1}{\theta_j^D} \left( \lambda_j^D - \sum_{k \in \mathcal{S}_j} m_{jk} \right) \text{ and}$$

$$q_k^{S,*}(\mathbf{m}) = \frac{1}{\theta_k^S} \left( \lambda_k^S - \sum_{j \in \mathcal{D}_k} m_{jk} \right).$$

The last formulas have an intuitive explanation: the expressions in the parentheses represent the expected



**Figure 4.** Average and Predicted Queue-Lengths

Notes. The solid lines represent the average queue lengths in the discrete-event stochastic model, and the dotted/triangle marker lines plot the functions  $q_1^{D,*}(\min\{\lambda_1^D, \lambda_1^S\})$  and  $q_1^{S,*}(\min\{\lambda_1^D, \lambda_1^S\})$  for a network with one demand node and one supply node. The parameters used are  $\lambda_1^D = 1$  per unit time and  $t = 100$ , and the patience time distributions for both nodes is the same.

demand (supply) that is not matched, and so to find the expected demand (supply) waiting to be matched, one needs to multiply by the mean patience time. The relationship is quadratic when the patience times are uniformly distributed in  $[0, \frac{2}{\theta_j^D}]$  and  $[0, \frac{2}{\theta_k^S}]$  and is

$$q_j^{D,*}(\mathbf{m}) = \frac{\lambda_j^D}{\theta_j^D} \left( 1 - \left( \frac{\sum_{k \in \mathcal{S}_j} m_{jk}}{\lambda_j^D} \right)^2 \right) \text{ and}$$

$$q_k^{S,*}(\mathbf{m}) = \frac{\lambda_k^S}{\theta_k^S} \left( 1 - \left( \frac{\sum_{j \in \mathcal{D}_k} m_{jk}}{\lambda_k^S} \right)^2 \right).$$

In some cases, the relationship cannot be written in a closed form, for instance, when the patience times follow a gamma distribution.

Despite (7) being nonconvex in general, we are able to characterize the structure of the objective function of (7) for a rather wide class of patience time distributions.

**Theorem 1.** *If the hazard rate functions of the patience time distributions are (strictly) increasing (decreasing), then  $q_j^{D,*}(\mathbf{m})$  and  $q_k^{S,*}(\mathbf{m})$  are (strictly) concave (convex) functions of the vector of matching rates  $\mathbf{m}$  for all  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ .*

**Corollary 1.** *If the hazard rate functions of the patience time distributions are increasing (decreasing), then the objective function of (7) is a convex (concave) function.*

Having introduced the MP and studied its properties, we are now ready to introduce our proposed matching policies.

### 3.2. A Matching Rate-Based Policy

Here, we define a policy that can mimic the optimal matching rates given by (7) as closely as possible

subject to the available stochastic demand and supply. Let  $\mathbf{m}$  be a feasible point of (7). Then, to mimic the matching rates defined by  $\mathbf{m}$ , let the amount of type  $j \in \mathbb{J}$  demand we match with type  $k \in \mathbb{K}$  supply for  $(j, k) \in \mathcal{E}$  in review period  $i \in \{1, \dots, \lfloor t/l \rfloor\}$  be

$$\mathcal{M}_{ijk}^r := \left\lfloor m_{jk} \min \left( l, \frac{Q_j^D(il-)}{\lambda_j^D}, \frac{Q_k^S(il-)}{\lambda_k^S} \right) \right\rfloor \geq 0, \quad (8)$$

which implies the cumulative number of matches made for  $(j, k) \in \mathcal{E}$  and  $t \geq 0$  is

$$M_{jk}^r(t) := \sum_{i=1}^{\lfloor t/l \rfloor} \mathcal{M}_{ijk}^r. \quad (9)$$

The matching rate-based policy is defined for each feasible point of (7) and for all patience time distributions and tries to achieve the feasible matching rates given by  $\mathbf{m}$ . The targeted rate during a review period between demand  $j \in \mathbb{J}$  and supply  $k \in \mathbb{K}$  is  $m_{jk}l$ . Further, the policy takes into account the fraction of available demand and supply matched, which ensures the queue lengths in (3) and (4) satisfy the nonnegativity condition.

**Proposition 1.** *The proposed matching policy given by (8) and (9) satisfies  $Q_j^D(t) \geq 0$  and  $Q_k^S(t) \geq 0$  for each  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$  and for all  $t \in \mathbb{R}_+$  for any feasible point  $\mathbf{m}$  to (7).*

As a result of Proposition 1, the matching rate-based policy satisfies the conditions given at the end of Section 2 for the associated matching process in (9) to be admissible. In particular, it is immediate to see that the matching process is integer-valued, nondecreasing, right-continuous with left limits, and does not use information about future arrivals to make matches (is nonanticipating).

The matching rate-based policy tries to mimic the matching rates at each edge and discrete review point by proportionally splitting the available demand (supply) at a node between the available supply (demand) at other nodes in the same proportions as the feasible matching rates in  $\mathbf{m}$ . Then, by using  $\mathbf{m}^*$  to define the amount matched in (8), we expect to achieve very close to the optimal matching rates (and we show that this is the case asymptotically in Theorem 4 in Section 5). However, the policy does not give any insight about the order in which edges should be prioritized for matching. Prioritizing edges can be useful for platforms wishing to implement simple rules for matching; that is, each demand (supply) type has a fixed ranking of the supply (demand) types that it checks for availability to match at each discrete review point. Such a static priority ordering is appealing because the ordering is state-independent and easy to describe to a practitioner. Thus, we want to understand when the matching rate-based policy reduces to a static priority ordering of the edges. In the next section, we show how to interpret an optimal extreme point solution as a static priority ordering of the edges.

### 3.3. A Priority-Ordering Policy

In this section, we propose a priority-ordering matching policy, which defines an order of edges over which matches are made in a greedy fashion at each discrete review point using available demand and supply. We show that, with a properly chosen priority ordering, such a policy is able to imitate the matching rates determined by an optimal extreme point solution to the MP when there exists one. From Corollary 1 to Theorem 1, the existence of an optimal extreme point solution is guaranteed when the patience time distributions have increasing hazard rates. In words, if supply and demand become more impatient the longer they wait in queue, then a simple priority ordering is enough to balance the trade-off between matching values and holding costs.

In the rest of this section, we assume the existence of an optimal extreme solution to the MP (7), which we denote by  $\mathbf{m}^*$ .

Before constructing the priority ordering, we need to establish a few properties of an optimal extreme point solution  $\mathbf{m}^*$ . It is helpful to define the following: we say a node  $j \in \mathbb{J}$  ( $k \in \mathbb{K}$ ) is slack if  $\sum_{k \in \mathbb{K}} m_{jk}^* < \lambda_j^D$  ( $\sum_{j \in \mathbb{J}} m_{jk}^* < \lambda_k^S$ ) and tight if  $\sum_{k \in \mathbb{K}} m_{jk}^* = \lambda_j^D$  ( $\sum_{j \in \mathbb{J}} m_{jk}^* = \lambda_k^S$ ); let  $E_0 = \{(j, k) \in \mathcal{E} : m_{jk}^* > 0\}$  denote the set of edges with positive matching rates, and let  $(\{\mathbb{J}, \mathbb{K}\}, E_0)$  denote the induced graph of  $\mathbf{m}^*$ , that is, the bipartite graph between node sets  $\mathbb{J}$  and  $\mathbb{K}$  with edges corresponding to positive

matching rates associated with  $\mathbf{m}^*$ . Moreover, a tree is a connected acyclic undirected graph, and a forest is an acyclic undirected graph; see Gross et al. (2018). Then, we have the following properties of the induced graph of an optimal extreme point solution.

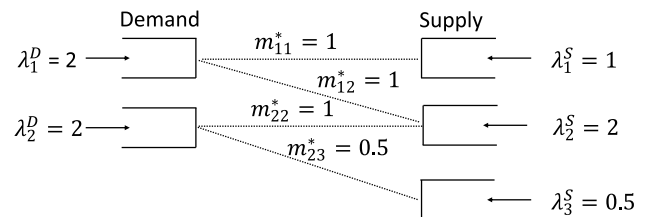
**Lemma 1.** Let  $\mathbf{m}^*$  be an optimal extreme point solution to (7). Then, the induced graph of  $\mathbf{m}^*$  has the following properties:

1. It has no cycles; that is, it is a forest or collection of trees.
2. Each distinct tree has at most one slack node.

With Lemma 1 in hand, we are now ready to describe the intuition for the priority ordering we construct. Lemma 1 guarantees that the set of edges,  $E_0$  on which  $\mathbf{m}^*$  makes matches can be partitioned into disjoint trees, each of which have at most one slack node (i.e., all nodes but at most one have their capacity used up by  $\mathbf{m}^*$ ). Intuitively, this allows us to construct the priority sets by iteratively following a path traversing the tight nodes of each tree, removing an edge at each iteration, and updating the remaining capacities by deducting the match that  $\mathbf{m}^*$  made on that edge. The edges removed in the  $l$ th iteration become the  $l$ th highest priority in our ordering; the process results in a smaller tree after each iteration and so works through all edges in finite time. This process continues until it reaches the final priority set that contains all the edges with  $m_{jk}^* = 0$ . Before we move to the general algorithm that constructs the priority sets, we present an example.

**Example 1.** Consider a system with two demand nodes and three supply nodes with arrival rates as depicted in Figure 5. The patience time distributions, values, and holding costs are such that an optimal extreme point solution to the MP for this system sets  $m_{11}^* = m_{12}^* = m_{22}^* = 1$  and  $m_{23}^* = 0.5$  as depicted in the figure with all other edges having  $m_{jk}^* = 0$  (e.g., this is the case if all matching values are one and all holding costs are zero). The edges with positive matching rate, that is,  $E_0$ , are drawn with dotted lines in the figure; note that these edges form a tree. For this optimal extreme point solution, we construct the following priority sets:  $\mathcal{P}_0(\mathbf{m}^*)$

**Figure 5.** A Matching Model with Two Demand and Three Supply Nodes



$= \{(1, 1), (2, 3)\}$ ,  $\mathcal{P}_1(\mathbf{m}^*) = \{(1, 2)\}$ ,  $\mathcal{P}_2(\mathbf{m}^*) = \{(2, 2)\}$ , and  $\mathcal{P}_3(\mathbf{m}^*) = \{(1, 3), (2, 1)\}$ . Observe that, in the first iteration, the algorithm selects two edges with tight leaf nodes from the tree (i.e., (1, 1) and (2, 3)) because they are disjoint, whereas in the second iteration, the algorithm picks only one of two available edges with tight leaf nodes (i.e., (1, 2) and (2, 2)) because they share the end point  $k=2$ , leaving the second edge until the next iteration, which is the last iteration processing edges with positive matching value. Note that, if edge (1, 2) ((2, 2)) were chosen in the first set  $\mathcal{P}_0(\mathbf{m}^*)$ , a priority policy would make too many matches along this edge and risk leaving agents at supply node 1 (3) unmatched. Then, the final step of the algorithm puts all edges with zero matching value in the set  $\mathcal{P}_3(\mathbf{m}^*)$ .

We formally present the construction of the priority sets in Algorithm 1, and here, we provide a more detailed explanation of the intuition of the algorithm. The order in which we traverse the trees in  $E_0$  is determined by the following observation. In each tree, at least one of the tight nodes must be a leaf of the tree; that is, it is connected to only one edge in  $E_0$  (or, equivalently, only one element of  $\mathbf{m}^*$  involving this node is positive). For an example, assume that  $j$  is such a leaf node in  $E_0$  with a tight constraint, that is,  $\sum_k m_{jk}^* = \lambda_j^D$ , and  $m_{jk'}^* > 0$  for some  $k'$ , whereas  $m_{jk}^* = 0$  for all other  $k \neq k'$ . Then, together, this implies that  $m_{jk'}^* = \lambda_j^D$ ; that is, there is only one edge,  $(j, k')$ , that uses up all of node  $j$ 's capacity. Our algorithm gives this edge highest priority because greedily matching all available supply and demand along this edge matches  $\min(\lambda_j^D, \lambda_{k'}^S) = \lambda_j^D$  (because we must have had  $\lambda_j^D \leq \lambda_{k'}^S$  for the initial value of  $m_{jk'}^*$  to be feasible), and this replicates the value of  $m_{jk'}^* = \lambda_j^D$ . Then, removing this edge and reducing the capacities by  $m_{jk'}^*$  results in a smaller tree with the same properties as  $E_0$ , so we can repeat the process until we work through the whole tree.

We note that Kerimov et al. (2023) establish existence of a similar priority ordering (for a model with no reneging) without explicitly constructing the priority sets as we do in Algorithm 1. Further, we make the following adjustment to allow for a potentially fewer number of priority sets in the final construction: more than one leaf node with a tight constraint can be put in the same priority set in a given iteration as long as the associated edges are disjoint (i.e., they do not share any endpoints).

Let  $H+1$  denote the final value of the counter  $h$  in Algorithm 1 so that  $\mathcal{P}_H(\mathbf{m}^*)$  denotes the last priority set with edges  $(j, k)$  such that  $m_{jk}^* > 0$ , and  $\mathcal{P}_{H+1}(\mathbf{m}^*)$  denotes the final priority set with all edges  $m_{jk}^* = 0$ .

#### Algorithm 1 (Priority Set Construction)

- 1: Input: supply, demand rates  $\lambda^D \in \mathbb{R}_+^J$ ,  $\lambda^S \in \mathbb{R}_+^K$ , and  $\mathbf{m}^*$  an optimal extreme point solution to (7)
- 2: Initialize priority set counter  $h=0$ , edge set  $E_0 = \{(j, k) \in \mathcal{E} : m_{jk}^* > 0\}$ , capacities  $\mathbf{d} = \lambda^D$ ,  $\mathbf{s} = \lambda^S$ , and  $E = E_0$
- 3: **while**  $|E| > 0$  **do**
- 4:   Initialize priority set:  $\mathcal{P}_h(\mathbf{m}^*) = \emptyset$
- 5:   Initialize consideration set (copy  $E$ ):  $C = E$
- 6:   **while**  $|C| > 0$  **do**
- 7:     Choose any  $(j, k) \in C$
- 8:     **if**  $m_{jk}^* = d_j$  or  $m_{jk}^* = s_k$  **then**
- 9:       Add to priority set:  $\mathcal{P}_h(\mathbf{m}^*) \leftarrow \mathcal{P}_h(\mathbf{m}^*) \cup \{(j, k)\}$
- 10:       Update capacities:  $s_k \leftarrow s_k - m_{jk}^*$ ,  $d_j \leftarrow d_j - m_{jk}^*$
- 11:       Remove  $(j, k)$  and neighbors:  $C \leftarrow C \setminus \{(j', k') \in C : j' = j \text{ or } k' = k\}$
- 12:     **else** Remove  $(j, k)$  from consideration:  $C \leftarrow C \setminus (j, k)$
- 13:     **end if**
- 14:   **end while**
- 15:   Remove priority set from edge set:  $E \leftarrow E \setminus \mathcal{P}_h$
- 16:   Increment counter:  $h \leftarrow h + 1$
- 17: **end while**
- 18: Last priority set:  $\mathcal{P}_h(\mathbf{m}^*) = \{(j, k) \in \mathcal{E} : m_{jk}^* = 0\}$ .

Note that Algorithm 1 returns nonoverlapping priority sets, and it terminates as the following result states.

**Lemma 2.** Algorithm 1 runs in finite time  $O(|E_0|^2)$ .

The sets  $\mathcal{P}_h(\mathbf{m}^*)$  form a partition of the edges,  $\mathcal{E}$ , of the bipartite graph. Let  $\mathcal{Q}_h(\mathbf{m}^*) = \bigcup_{l=0}^h \mathcal{P}_l(\mathbf{m}^*)$  denote the set of all priority edges chosen up to iteration  $h$ . Given this partition, define the following solution to (7) recursively for  $(j, k) \in \mathcal{P}_h(\mathbf{m}^*)$  as

$$y_{jk}^p(\mathbf{m}^*) = \min \left( \lambda_j^D - \sum_{k': (j, k') \in \mathcal{Q}_{h-1}(\mathbf{m}^*)} y_{jk'}^p(\mathbf{m}^*), \lambda_k^S - \sum_{j': (j', k) \in \mathcal{Q}_{h-1}(\mathbf{m}^*)} y_{j'k}^p(\mathbf{m}^*) \right),$$

where an empty sum is defined to be zero.

**Proposition 2.** For  $\mathbf{m}^*$  an optimal extreme point solution to (7), we have  $y_{jk}^p(\mathbf{m}^*) = m_{jk}^*$  for all  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ .

We are ready now to define recursively the priority-ordering matching policy. The amount of type  $j \in \mathbb{J}$  demand we match with type  $k \in \mathbb{K}$  supply for  $(j, k) \in \mathcal{P}_0$  at the end of review period  $i \geq 1$  is

$$\mathcal{M}_{ijk}^p := \min(Q_j^D(i-), Q_k^S(i-)). \quad (10)$$



For  $(j, k) \in \mathcal{P}_h$ ,  $h = 1, \dots, JK$  and  $i \in \{1, \dots, \lfloor t/l \rfloor\}$ , define recursively

$$\mathcal{M}_{ijk}^p := \min \left( Q_j^D(il-) - \sum_{k': (j, k') \in \mathcal{Q}_{h-1}(m^*)} \mathcal{M}_{ijk'}^p, Q_k^S(il-) - \sum_{j': (j', k) \in \mathcal{Q}_{h-1}(m^*)} \mathcal{M}_{ij'k}^p \right). \quad (11)$$

The aggregate number of matches at time  $t \geq 0$  is given by

$$M_{jk}^p(t) := \sum_{i=1}^{\lfloor t/l \rfloor} \mathcal{M}_{ijk}^p. \quad (12)$$

It is straightforward to see that the priority-ordering matching policy satisfies the admissibility conditions given at the end of Section 2.

The implementation of the priority-ordering policy does not require knowledge of the optimal extreme point solution once the priority sets are constructed. In other words, a computer program needs to know only the priority sets to be able to implement the aforementioned policy.

**Remark 1** (Connection to the  $\frac{c\mu}{\theta}$  Rule). The priority ordering studied in this section can be connected to the well-known  $\frac{c\mu}{\theta}$  rule studied for multiclass queueing systems in which  $\mu$  denotes the service rate (Smith 1956, Atar et al. 2011). To see this, observe that, for a system with a single supply node and exponential patience time distributions, the objective function of the MP takes the form:  $\sum_j a_j m_{j1}$ , where  $a_j = v_{j1} + \frac{c_j^D}{\theta_j^D} + \frac{c_j^S}{\theta_j^S}$ . Hence, its optimal solution is given by a priority rule that orders the weights  $a_j$ . These weights do not depend on the arrival rates and are similar to that of the  $\frac{c\mu}{\theta}$  rule but are modified to include the matching values and the fact that there is not service rate in our model. However, in the general case, the priority ordering cannot be written as a simple priority rule because of the dimension of the model and the general patience time distributions.

We propose matching policies that use the information of an optimal solution to the MP, which involves an approximation of the demand and supply queue lengths. The study of the asymptotic performance of the aforementioned policies requires showing a connection between (1), which involves the stochastic queue lengths, and the objective of the MP in (7). We do this by developing and analyzing a fluid model that approximates the demand and supply queue lengths in Section 4. We then use that analysis in Section 5 to develop an asymptotic connection between (1) and (7).

## 4. The Fluid Model and Its Convergence to Invariant States

Section 4.1 presents a fluid model that can be seen as an approximation to the discrete-event stochastic model developed in Section 2. The invariant states of the fluid model are fixed points of the fluid model equations, and their characterization is required to develop the proposed matching policies in Section 3. Section 4.2 characterizes the fluid model invariant states and shows that a fluid model solution converges to an invariant state as time becomes large. The fluid model presented in Section 4.1 is exactly the fluid model presented in section 3.1 in Aveklouris et al. (2023) except restricted to linear arrival functions and linear matching functions.

In this section, we require the following notation. Given a Polish space  $\mathbb{S}$ , we use the notation  $\mathcal{C}(\mathbb{S})$  (with no subscript) to denote the set of  $\mathbb{S}$ -valued functions with domain  $\mathbb{R}_+$  that are continuous. We endow  $\mathcal{C}(\mathbb{S})$  with the usual Skorokhod  $J_1$ -topology (Billingsley 1999). Recall that, for  $L \in [0, \infty]$ ,  $\mathcal{M}[0, L]$  denotes the set of finite, nonnegative Borel measures on  $[0, L]$  endowed with the topology of weak convergence, which is a Polish space. Given a measure  $\nu \in \mathcal{M}[0, L]$ , we write  $\nu[0, x] := \int_0^x \nu(dy)$ . Given a measure  $\nu \in \mathcal{M}[0, L]$  and a Borel measurable function  $f: [0, L] \rightarrow \mathbb{R}$  that is integrable with respect to  $\nu$ , define  $\langle f, \nu \rangle := \int_{[0, L]} f(x) \nu(dx)$ .

### 4.1. The Fluid Model

The fluid model state space parallels the state space for the discrete-event stochastic model in Section 2 except that we do not need to track the times elapsed since the last arrival of each agent type (because arrivals occur continuously in time). Recall that  $H_j^D$  and  $H_k^S$  are the right edges of the support of the patience time distribution functions for any  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ . The fluid model state at time  $t \in \mathbb{R}_+$  is described as follows: for each  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ ,

- $\bar{Q}_j^D(t)$  and  $\bar{Q}_k^S(t)$  are the fluid queue lengths at time  $t$ .
- $\bar{\eta}_j^D(t) \in \mathcal{M}[0, H_j^D]$  and  $\bar{\eta}_k^S(t) \in \mathcal{M}[0, H_k^S]$  are measure-valued functions that provide upper bounds on the fluid queue lengths at every time  $t$ ; that is, analogous to (2),  
 $\bar{Q}_j^D(t) \leq \langle 1, \bar{\eta}_j^D(t) \rangle$ , for all  $j \in \mathbb{J}$ , and  $\bar{Q}_k^S(t) \leq \langle 1, \bar{\eta}_k^S(t) \rangle$ ,  
for all  $k \in \mathbb{K}$ .

(13)

In summary, a fluid model solution is a vector  $(\bar{Q}^D, \bar{Q}^S, \bar{\eta}^D, \bar{\eta}^S) \in \mathcal{C}(\bar{\mathbb{Y}})$  having components  $(\bar{Q}_j^D(t), j \in \mathbb{J})$ ,  $(\bar{Q}_k^S(t), k \in \mathbb{K})$ ,  $(\bar{\eta}_j^D(t), j \in \mathbb{J})$ , and  $(\bar{\eta}_k^S(t), k \in \mathbb{K})$  that take values in the subset  $\bar{\mathbb{Y}}$  of

$$\mathbb{R}_+^J \times \mathbb{R}_+^K \times \times_{j=1}^J \mathbf{M}[0, H_j^D] \times \times_{k=1}^K \mathbf{M}[0, H_k^S],$$

for which (13) holds for each  $t \in \mathbb{R}_+$ . We require that a fluid model solution  $(\bar{Q}^D, \bar{Q}^S, \bar{\eta}^D, \bar{\eta}^S) \in \mathcal{C}(\bar{\mathbb{Y}})$  satisfies



finiteness conditions such that, for all  $t \geq 0$ ,

$$\begin{aligned} \int_0^t \langle h_j^D, \bar{\eta}_j^D(u) \rangle du &< \infty, \text{ for all } j \in \mathbb{J}, \text{ and} \\ \int_0^t \langle h_k^S, \bar{\eta}_k^S(u) \rangle du &< \infty, \text{ for all } k \in \mathbb{K}, \end{aligned} \quad (14)$$

and has initial potential queue measures with no atoms; that is,

$$\begin{aligned} \langle 1_{\{x\}}, \bar{\eta}_j^D(0) \rangle &= 0 \text{ for all } x \in [0, H_j^D], j \in \mathbb{J} \text{ and} \\ \langle 1_{\{x\}}, \bar{\eta}_k^S(0) \rangle &= 0 \text{ for all } x \in [0, H_k^S], k \in \mathbb{K}. \end{aligned} \quad (15)$$

The fluid analogues of the cumulative reneging processes are explained in words subsequently and are, for  $t \geq 0$  and each  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ ,

$$\bar{R}_j^D(t) = \int_0^t \int_0^{H_j^D} h_j^D(x) 1_{\{\bar{\eta}_j^D(u)[0,x] < \bar{Q}_j^D(u)\}} \bar{\eta}_j^D(u)(dx)du \quad (16)$$

and

$$\bar{R}_k^S(t) = \int_0^t \int_0^{H_k^S} h_k^S(x) 1_{\{\bar{\eta}_k^S(u)[0,x] < \bar{Q}_k^S(u)\}} \bar{\eta}_k^S(u)(dx)du. \quad (17)$$

The inner integrals in (16) and (17) represent the instantaneous reneging rate, which is determined by the hazard rate function and fluid age. Then, integrating over the instantaneous reneging rate in  $[0, t]$  gives the cumulative reneging up to time  $t$ . When the patience times are exponentially distributed with rates  $\theta_j^D$  and  $\theta_k^S$  so that the  $h_j^D(x) = \theta_j^D$  and  $h_k^S(x) = \theta_k^S$  for  $x \in \mathbb{R}_+$ , we have that the instantaneous rate at which fluid leaves without being matched is linear in the queue length; that is,

$$\bar{R}_j^D(t) = \int_0^t \theta_j^D \bar{Q}_j^D(u) du \text{ and } \bar{R}_k^S(t) = \int_0^t \theta_k^S \bar{Q}_k^S(u) du.$$

Condition (15) ensures that  $\bar{R}_j^D(t)$  and  $\bar{R}_k^S(t)$  in (16) and (17) are finite for all  $t \in \mathbb{R}_+$ .

The measure-valued functions  $\bar{\eta}_j^D \in \mathcal{M}[0, H_j^D]$  and  $\bar{\eta}_k^S \in \mathcal{M}[0, H_k^S]$  must satisfy the equations given subsequently and interpreted in words after for given input to the fluid model  $\lambda^D \in (0, \infty)^J$  and  $\lambda^S \in (0, \infty)^K$ . We interpret  $\lambda_j^D$  as the type  $j \in \mathbb{J}$  fluid arrival rate and  $\lambda_k^S$  as the type  $k \in \mathbb{K}$  fluid arrival rate, meaning that  $\lambda_j^D t$  and  $\lambda_k^S t$  are the cumulative amounts of type  $j$  and  $k$  fluid to arrive by time  $t \in \mathbb{R}_+$ . For any bounded function  $f \in \mathcal{C}(\mathbb{R}_+)$  the following integral equations hold for each  $j \in \mathbb{J}$ ,  $k \in \mathbb{K}$ , and  $t \geq 0$ ,

$$\begin{aligned} \langle f, \bar{\eta}_j^D(t) \rangle &= \int_0^{H_j^D} f(x+t) \frac{1 - G_j^D(x+t)}{1 - G_j^D(x)} \bar{\eta}_j^D(0)(dx) \\ &+ \lambda_j^D \int_0^t f(t-u)(1 - G_j^D(t-u))du, \end{aligned} \quad (18)$$

and

$$\begin{aligned} \langle f, \bar{\eta}_k^S(t) \rangle &= \int_0^{H_k^S} f(x+t) \frac{1 - G_k^S(x+t)}{1 - G_k^S(x)} \bar{\eta}_k^S(0)(dx) \\ &+ \lambda_k^S \int_0^t f(t-u)(1 - G_k^S(t-u))du. \end{aligned} \quad (19)$$

The first term of the integral equations for measures  $\bar{\eta}_j^D(\cdot)$  and  $\bar{\eta}_k^S(\cdot)$  in (18) and (19) tracks when the patience time of fluid demand/supply initially present in the system at time zero expires, and the second term has an analogous meaning for the newly arriving fluid.

The specification of a fluid model solution for given arrival rates  $\lambda^D \in (0, \infty)^J$  and  $\lambda^S \in (0, \infty)^K$  and an initial condition requires the specification of the matching rates, which must lie in the set

$$\begin{aligned} \mathbb{M} := \left\{ \mathbf{m} \in \mathbb{R}_+^{J \times K} : \sum_{k \in \mathcal{S}_j} m_{jk} \leq \lambda_j^D, \sum_{j \in \mathcal{D}_k} m_{jk} \leq \lambda_k^S, \right. \\ \left. \text{and } m_{jk} = 0 \quad \forall (j, k) \notin \mathcal{E} \right\}. \end{aligned}$$

For a given matching rate matrix  $\mathbf{m} \in \mathbb{M}$ ,  $m_{jk}$  is the rate at which type  $j \in \mathbb{J}$  customer fluid is matched with type  $k \in \mathbb{K}$  worker fluid so that  $m_{jk} t$  is the cumulative amount of fluid of those types matched by time  $t \in \mathbb{R}_+$ . Then, the fluid queue lengths evolve as follows: for all  $j \in \mathbb{J}$ ,  $k \in \mathbb{K}$  and  $t \geq 0$ ,

$$\bar{Q}_j^D(t) = \bar{Q}_j^D(0) + \lambda_j^D t - \bar{R}_j^D(t) - \sum_{k \in \mathcal{S}_j} m_{jk} t, \quad (20)$$

and,

$$\bar{Q}_k^S(t) = \bar{Q}_k^S(0) + \lambda_k^S t - \bar{R}_k^S(t) - \sum_{j \in \mathcal{D}_k} m_{jk} t. \quad (21)$$

Equations (20) and (21) are the fluid analogues of the queue-length evolution Equations (3) and (4) in the stochastic model.

**Definition 1.** Let  $\lambda^D \in (0, \infty)^J$  and  $\lambda^S \in (0, \infty)^K$  be given. A fluid model solution for arrival rates  $(\lambda^D, \lambda^S)$  is  $(\bar{Q}^D, \bar{Q}^S, \bar{\eta}^D, \bar{\eta}^S) \in \mathcal{C}(\mathbb{Y})$  that satisfies Conditions (14) and (15) and the integral equations (18) and (19) and is such that there exists a matching rate matrix  $\mathbf{m} \in \mathbb{M}$  for which (20) and (21) hold with  $\bar{R}^D$  and  $\bar{R}^S$  given by (16) and (17).

**Remark 2.** A fluid model solution for given arrival rates  $(\lambda^D, \lambda^S)$  and the given initial state is unique; see theorem 1 in Aveklouris et al. (2023).

## 4.2. Invariant Analysis

The matching problem (7) that approximates the objective (1) to maximize the long-run average matching value in the discrete event stochastic system follows by identifying the invariant states of the fluid model. In particular, the approximations for the mean steady-state queue lengths used in (7) are invariant states of the fluid model.

**Definition 2.** Let  $\lambda^D \in (0, \infty)^J$  and  $\lambda^S \in (0, \infty)^K$ . A tuple  $(q^{D,*}, q^{S,*}, \eta^{D,*}, \eta^{S,*}) \in \bar{\mathbb{Y}}$  is an invariant state for  $(\lambda^D, \lambda^S)$  if the constant function  $(\bar{Q}^D, \bar{Q}^S, \bar{\eta}^D, \bar{\eta}^S)$  given by

$$(\bar{Q}^D(t), \bar{Q}^S(t), \bar{\eta}^D(t), \bar{\eta}^S(t)) = (q^{D,*}, q^{S,*}, \eta^{D,*}, \eta^{S,*}), \quad \text{for all } t \geq 0 \quad (22)$$

is a fluid model solution for  $(\lambda^D, \lambda^S)$ . The invariant manifold for  $\lambda = (\lambda^D, \lambda^S)$  is the set of all invariant states for  $(\lambda^D, \lambda^S)$ , which we denote by  $\mathcal{I}_\lambda$ .

Recall that  $G_{e,j}^D(x)$ ,  $G_{e,k}^S(x)$  for  $j \in \mathbb{J}, k \in \mathbb{K}, x \in \mathbb{R}_+$  are the excess life distributions of the patience times.

**Proposition 3.** Let  $\lambda^D \in (0, \infty)^J$  and  $\lambda^S \in (0, \infty)^K$ . For  $m \in \mathbb{M}$ , define for each  $x \in \mathbb{R}_+$ ,  $j \in \mathbb{J}$ , and  $k \in \mathbb{K}$ :

$$\eta_j^{D,*}(dx) := \lambda_j^D (1 - G_j^D(x)) dx, \quad (23)$$

$$\eta_k^{S,*}(dx) := \lambda_k^S (1 - G_k^S(x)) dx, \quad (24)$$

$$q_j^{D,*}(m) := \begin{cases} \frac{\lambda_j^D}{\theta_j^D}, & \text{if } \sum_{k \in S_j} m_{jk} = 0, \\ \frac{\lambda_j^D}{\theta_j^D} G_{e,j}^D \left( (G_j^D)^{-1} \left( 1 - \frac{\sum_{k \in S_j} m_{jk}}{\lambda_j^D} \right) \right), & \text{if } \sum_{k \in S_j} m_{jk} \in (0, \lambda_j^D], \end{cases} \quad (25)$$

$$q_k^{S,*}(m) := \begin{cases} \frac{\lambda_k^S}{\theta_k^S}, & \text{if } \sum_{j \in D_k} m_{jk} = 0, \\ \frac{\lambda_k^S}{\theta_k^S} G_{e,k}^S \left( (G_k^S)^{-1} \left( 1 - \frac{\sum_{j \in D_k} m_{jk}}{\lambda_k^S} \right) \right), & \text{if } \sum_{j \in D_k} m_{jk} \in (0, \lambda_k^S]. \end{cases} \quad (26)$$

Then,  $(q^{D,*}(m), q^{S,*}(m), \eta^{D,*}, \eta^{S,*}) \in \bar{\mathbb{Y}}$  is in the invariant manifold  $\mathcal{I}_\lambda$  for any  $m \in \mathbb{M}$ . Conversely, given  $m \in \mathbb{M}$ , if  $(q^D(m), q^S(m), \eta^D, \eta^S) \in \mathcal{I}_\lambda$ , then

$$(q^D(m), q^S(m), \eta^D, \eta^S) = (q^{D,*}(m), q^{S,*}(m), \eta^{D,*}, \eta^{S,*}).$$

Proposition 3 shows that there is a nonlinear relationship between the matching rate matrix  $m \in \mathbb{M}$  and the invariant fluid queue lengths that depends on the patience time distributions. This is not surprising given the nonlinear relationship between the fluid queue

lengths and the patience time distribution in many server queues with abandonment (see theorem 3.1 in Whitt 2006, theorem 5.5 in Kang and Ramanan 2012, theorem 3.2 in Zhang 2013, theorem 1 in Puha and Ward 2019).

The long-time behavior of the fluid model is characterized by the invariant manifold identified in Proposition 3.

**Theorem 2.** For each  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ , assume  $h_j^D$  and  $h_k^S$  are bounded functions. Suppose that  $\lambda^D \in (0, \infty)^J$  and  $\lambda^S \in (0, \infty)^K$ , and  $(\bar{Q}^D, \bar{Q}^S, \bar{\eta}^D, \bar{\eta}^S) \in \mathcal{C}(\bar{\mathbb{Y}})$  is a fluid model solution for  $(\lambda^D, \lambda^S)$ . Then,<sup>4</sup>

$$\lim_{t \rightarrow \infty} (\bar{Q}^D(t), \bar{Q}^S(t), \bar{\eta}^D(t), \bar{\eta}^S(t)) = (q^{D,*}(m), q^{S,*}(m), \eta^{D,*}, \eta^{S,*}). \quad (27)$$

## 5. Performance Analysis

In this section, we study the performance of the proposed matching policies in a high-volume setting when the goal is to optimize (1). We first derive an upper bound on the objective function in a high-volume setting in Section 5.1 and then show that the matching rate-based and priority-ordering policies achieve this upper bound asymptotically in Section 5.2.

Throughout this section, we assume the system is initially empty. To accommodate more general initial conditions, we replicate assumption 3 in Aveklouris et al. (2023) and then observe that the fluid model in Section 4 accommodates all initial conditions allowed in that assumption. We chose not to present assumption 3 in Aveklouris et al. (2023) here and, instead, assume an empty initial system to eliminate the notational overhead.

### 5.1. High-Volume Setting

We consider a high-volume setting in which the arrival rates of demand and supply grow large. Specifically, we consider a family of systems indexed by  $n \in \mathbb{N}$ , where  $n$  tends to infinity, and let the arrival rates in the  $n$ th system be

$$\lambda_j^{D,n} = n\lambda_j^D \text{ for all } j \in \mathbb{J} \text{ and } \lambda_k^{S,n} = n\lambda_k^S \text{ for all } k \in \mathbb{K}. \quad (28)$$

Otherwise, each system  $n$  has the same basic structure as that of the system described in Section 2. Note that the patience time distributions do not scale with  $n$ , which is consistent with the existing literature; see, for example, Kang and Ramanan (2010).

We append a superscript  $n$  to the system processes associated with the system having arrival rates as given in (28). Then,  $M^n$  is the matching process associated with the system having arrival rates in (28), and  $V_{M^n}$  is the objective value in (1) under  $M^n$ . Also,

$(\alpha^{D,n}, \alpha^{S,n}, Q^{D,n}, Q^{S,n}, \eta^{D,n}, \eta^{S,n})$  is the associated state process.

In the high-volume setting, the fluid model given in Section 4 serves as an approximation to the discrete-event stochastic model given in Section 2, and the large time behavior of the fluid model established in Theorem 2 motivates the construction of the MP in (7) in Section 3. Hence, in high volume, we expect that the optimal objective value of the MP provides an upper bound on the value of any admissible policy given in (1), and the next result validates this intuition. Note that the objective function value should be of order  $n$  because arrival rates are of order  $n$ , and so the objective function in the following is multiplied by  $1/n$ .

**Theorem 3.** For each  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ , assume  $h_j^D$  and  $h_k^S$  are bounded functions. Let  $m^*$  be an optimal solution to (7). If  $\{M^n\}_{n \in \mathbb{N}}$  is a sequence of matching processes such that, as  $n \rightarrow \infty$ ,

$$\sup_{0 \leq t \leq T} \left| \frac{M_{jk}^n(t)}{n} - m_{jk}t \right| \rightarrow 0, \quad (29)$$

in probability for some  $m \in \mathbb{M}$  and for all  $T \geq 0$ , then

$$\begin{aligned} & \lim_{t \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{1}{nt} V_{M^n}(t) \\ &= \sum_{(j,k) \in \mathcal{E}} v_{jk} m_{jk} - \sum_{j \in \mathbb{J}} c_j^D q_j^{D,*}(m) - \sum_{k \in \mathbb{K}} c_k^S q_k^{S,*}(m) \\ &\leq \sum_{(j,k) \in \mathcal{E}} v_{jk} m_{jk}^* - \sum_{j \in \mathbb{J}} c_j^D q_j^{D,*}(m^*) - \sum_{k \in \mathbb{K}} c_k^S q_k^{S,*}(m^*). \quad (30) \end{aligned}$$

The condition in Theorem 3 ensures that the scaled cumulative matches made build up linearly in time, which is an assumption in the fluid model Equations (20) and (21).

Having established an upper bound, we investigate the existence of a policy that achieves this upper bound in the high-volume setting.

**Definition 3.** Let  $m^*$  be an optimal solution to (7). When  $h_j^D$  and  $h_k^S$  are bounded functions for each  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ , an admissible policy  $M^n(\cdot)$  is asymptotically optimal if

$$\lim_{t \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{1}{nt} \frac{V_{M^n}(t)}{\sum_{(j,k) \in \mathcal{E}} v_{jk} m_{jk}^* - \sum_{j \in \mathbb{J}} c_j^D q_j^{D,*}(m^*) - \sum_{k \in \mathbb{K}} c_k^S q_k^{S,*}(m^*)} = 1,$$

in probability.

In the remainder of Section 5, we study the asymptotic behavior of our proposed discrete review

matching policies. To do so, we also need to scale the review period length as follows:

$$l^n = \frac{1}{n^{2/3}} l.$$

The intuition behind the scaling of the length of the review period is that it should be long enough to allow us to make the most valuable matches as dictated by an MP optimal solution but also short enough that the renegeing does not hurt. The scaling of the review period length is not unique and is connected to the number of moments assumed on the interarrival times. The assumption that the interarrival time random variables have a finite fifth moment is used to bound the number of arrivals that occur during each review period.<sup>5</sup>

## 5.2. Asymptotic Optimality of the Matching Policies

The goal of this section is to show that the proposed matching policies defined in Section 3 are asymptotically optimal. We start the analysis by examining the matching rate-based policy that is introduced in Section 3.2.

Given a feasible point  $m$  to (7) and  $t \geq 0$ , the number of matches for the matching rate-based policy in the  $n$ th system at discrete review period  $i \in \{1, \dots, \lfloor t/l^n \rfloor\}$  is given by

$$\begin{aligned} \mathcal{M}_{ijk}^{r,n} &:= \left\lfloor nm_{jk} \min \left( l^n, \frac{Q_j^{D,n}(il^n -)}{\lambda_j^{D,n}}, \frac{Q_k^{S,n}(il^n -)}{\lambda_k^{S,n}} \right) \right\rfloor \\ &= \left\lfloor m_{jk} \min \left( n^{1/3} l, \frac{Q_j^{D,n}(il^n -)}{\lambda_j^D}, \frac{Q_k^{S,n}(il^n -)}{\lambda_k^S} \right) \right\rfloor. \quad (31) \end{aligned}$$

Note that the instantaneous matching rates  $m_{jk}$  are scaled by  $n$  in (31) because they remain feasible to (7) when the arrival rates are scaled by a factor of  $n$ . The cumulative number of matches until time  $t$  in the  $n$ th system is given by

$$M_{jk}^{r,n}(t) := \sum_{i=1}^{\lfloor t/l^n \rfloor} \mathcal{M}_{ijk}^{r,n}. \quad (32)$$

Our goal is to show that the matching rate-based policy is asymptotically optimal. The following result shows that the scaled number of matches approaches asymptotically any feasible point of MP, and it is one of the keys to prove the asymptotic optimality.

**Theorem 4.** Let  $m$  be a feasible point to (7). Under the matching rate-based policy (31) and (32), as  $n \rightarrow \infty$ ,

$$\sup_{0 \leq t \leq T} \left| \frac{M_{jk}^{r,n}(t)}{n} - m_{jk}t \right| \rightarrow 0,$$

in probability for all  $T \geq 0$ ,  $(j, k) \in \mathcal{E}$ .

Although the detailed proof of Theorem 4 is given in the online appendix, we provide the reader with a brief

outline here to establish the basic template used in the proofs of this section. The proof proceeds in three basic steps: (i) establish an upper bound on the amount of renegeing during a review period, (ii) use the renegeing upper bound to derive a lower bound on the number of matches made during a review period, and (iii) show that this lower bound on matches made approaches the desired feasible point of the MP.

Theorem 4 holds for an optimal solution  $\mathbf{m}^*$  to (7) and guarantees that the condition required in Theorem 3 is satisfied. Moreover, as shown in the proof of Theorem 3, theorems 1 and 3 in Aveklouris et al. (2023) establish that a fluid model solution arises as a limit point of the fluid scaled state descriptors<sup>6</sup> ( $Q^{D,n}/n, Q^{S,n}/n, \eta^{D,n}/n, \eta^{S,n}/n$ ), and Theorem 2 ensures that that fluid model solution approaches the invariant point defined in Proposition 3 for  $\mathbf{m}^*$ . We conclude that the matching rate-based policy is asymptotically optimal.

**Corollary 2.** For each  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ , assume  $h_j^D$  and  $h_k^S$  are bounded functions. The matching rate-based policy is asymptotically optimal for an optimal solution,  $\mathbf{m}^*$ , to (7).

Having studied the asymptotic performance of the matching rate-based policy, we now move to the asymptotic optimality of the priority-ordering policy introduced in Section 3.3. In this case, the number of matches made in the  $n$ th system between type  $j \in \mathbb{J}$  demand and type  $k \in \mathbb{K}$  supply for  $(j, k) \in \mathcal{P}_h, h = 0, 1, \dots, H+1$  at the end of review period  $i \in \{1, \dots, \lfloor t/l^n \rfloor\}$  is given by

$$\mathcal{M}_{ijk}^{p,n} := \min \left( Q_j^{D,n}(il^n) - \sum_{k': (j, k') \in \mathcal{Q}_{h-1}(\mathbf{m}^*)} \mathcal{M}_{ijk'}^{p,n}, \right. \\ \left. Q_k^{S,n}(il^n) - \sum_{j': (j', k) \in \mathcal{Q}_{h-1}(\mathbf{m}^*)} \mathcal{M}_{ij'k}^{p,n} \right), \quad (33)$$

where an empty sum is defined to be zero. Hence, the aggregate number of matches at time  $t \geq 0$  is given by

$$M_{jk}^{p,n}(t) := \sum_{i=1}^{\lfloor t/l^n \rfloor} \mathcal{M}_{ijk}^{p,n}. \quad (34)$$

The following result states that the priority-ordering policy asymptotically achieves the optimal matching rates, and its corollary, which is analogous to Corollary 2 to Theorem 4, shows that the priority-ordering policy is asymptotically optimal.

**Theorem 5.** Suppose that there exists an optimal extreme point solution to (7) denoted by  $\mathbf{m}^*$  and let the priority sets be given by Algorithm 1. Under the priority-ordering matching policy (33) and (34), as  $n \rightarrow \infty$ ,

$$\sup_{0 \leq t \leq T} \left| \frac{M_{jk}^{p,n}(t)}{n} - m_{jk}^* t \right| \rightarrow 0,$$

in probability for all  $T \geq 0, (j, k) \in \mathcal{E}$ .

The proof of the last theorem is based on a two-level induction. We first proceed in an induction on the priority sets, and then for any set, we proceed in a second induction on the review periods inside this set. For each review period, we show that the number of matches made is close to the optimal matching rates given by the MP. Now, the asymptotic optimality of the priority-ordering policy follows in the same way as in the matching rate-based policy.

**Corollary 3.** For each  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ , assume  $h_j^D$  and  $h_k^S$  are bounded functions. The priority-ordering policy is asymptotically optimal for an optimal extreme point solution,  $\mathbf{m}^*$ , to (7).

Theorems 4 and 5 suggest that the quantity  $m_{jk}^* t$  can be seen as an approximation of the mean number of matches at time  $t$  for  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ .

**Remark 3.** The matching rate-based policy and priority-ordering policy can be defined for any patience time distributions that have a density, a finite mean, and a strictly increasing cdf because these are the requirements needed to define the fluid model in Definition 1 and its invariant states in Proposition 3. Even though Corollary 2 to Theorem 4 and Corollary 3 to Theorem 5 require bounded hazard functions, we can still simulate the performance of the policies when the hazard functions are unbounded (as, for example, is true for the uniform distribution), and we do this in Section 6.

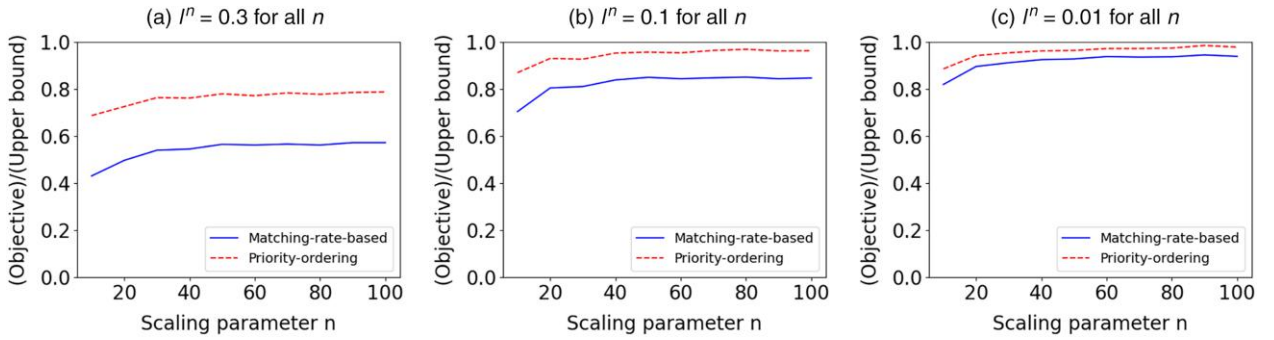
## 6. Simulation Experiments

In this section, we present simulation results on the behavior of the aforementioned matching policies in matching models with small numbers of nodes that have nontrivial behavior. We consider the objective (1) of the matching policies and illustrate its behavior in several different parameter regimes. Specifically, we consider two broad sets of simulations; the first assesses policy performance in a network as the scaling parameter  $n$  grows, the second investigates how to practically set the review period length in a system for a fixed  $n$ .

### 6.1. The Effect of Increasing Arrival Rates

For our first set of simulations, we consider a model with four demand and four supply nodes and Poisson arrivals with rates  $\lambda^D = (3, 2, 1, 3)$  and  $\lambda^S = (2, 2, 2, 2)$ . Further, we assume that the system is initially empty, and we denote the value vector for each demand node by  $\mathbf{v}_j = (v_{j1}, \dots, v_{jK})$  for each  $j \in \mathbb{J}$ . The values at each edge are given by the vectors:  $\mathbf{v}_1 = (1, 2, 3, 1)$ ,  $\mathbf{v}_2 = (1, 1, 1, 1)$ ,  $\mathbf{v}_3 = (2, 1, 1, 2)$ , and  $\mathbf{v}_4 = (3, 3, 2, 1)$ . The holding costs for demand and supply nodes are  $\mathbf{c}^D = (1, 2, 1, 2)$  and  $\mathbf{c}^S = (2, 1, 2, 1)$ , respectively. All the nodes have the same patience time distribution with the same mean and variance, that is,  $\theta_j^D = \theta_k^S = \theta$ . We consider two



**Figure 6.** (Color online) Gamma Distributed Patience Times with Mean  $1/3$  and Variance  $1/27$ 

different distributions of the patience times both having an increasing hazard rate function; uniform in  $[0, 2/\theta]$  and gamma with shape parameter equal to three and scale parameter equal to  $1/3\theta$ , that is, the mean patience time is  $1/\theta$ . We fix the time horizon  $t = 100$  in (1). Figures 6 and 7 show the ratio of the average objective of the matching policy to the optimal objective of (7) for the two policies as a function of the scaling parameter  $n$  for various values of the discrete review period  $l^n$ . To illustrate the impact of changing both the review period length,  $l^n$ , and the scaling parameter,  $n$ , in each chart of this section we hold the review period length  $l^n$  constant, letting the scaling parameter  $n$  grow (i.e., we do not let the review period length grow with the arrival rate as it did in Section 5.1). In other words, in each chart, we hold the discrete review length constant, letting the arrival rates  $\lambda^{D,n}$  and  $\lambda^{S,n}$  grow large. We do this because our chosen scaling for  $l^n$  is not unique, and we want to illustrate more broadly the review period length impact. We next summarize the main observations from the simulations.

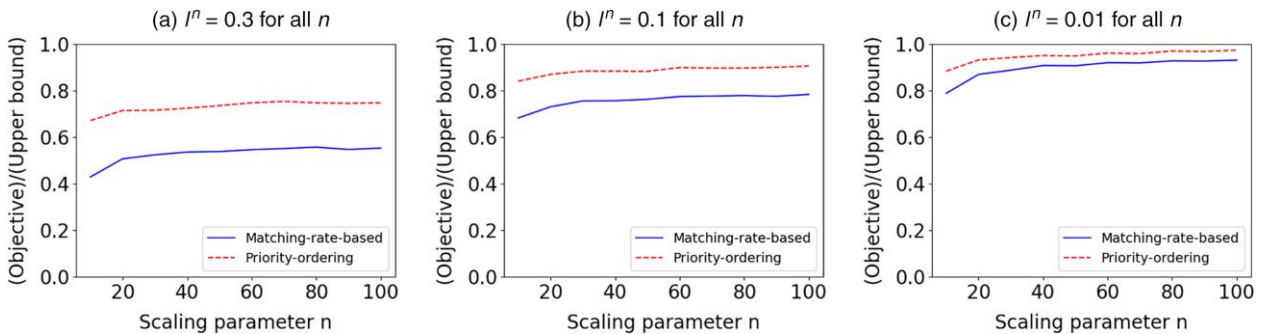
The matching policies have better performance for small review period length. Figures 6 and 7 show that the policies perform better (are much closer to the upper bound) when the review period length is small, a phenomenon we explore further in the simulations of the next section. Moreover, Figures 6(c) and 7(c) show

that the policy performance can be good for small arrival rates even though the asymptotic optimality results (Theorems 4 and 5) hold for large arrival rates.

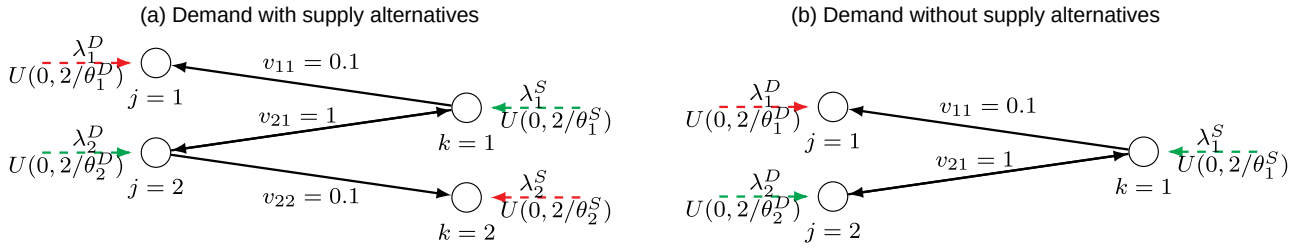
The priority-ordering policy seems to perform better than the matching rate-based policy. In all figures, the priority-ordering policy behaves better than the matching rate-based policy. This can be explained intuitively as follows: First, the priority-ordering policy makes matches at each review period according to the priority of the edges that identifies the most valuable edges taking into account the holding costs as well. Second, the priority-ordering policy exhausts the demand or supply at each review period in contrast to the matching rate-based policy. Based on this observation, we focus on the priority-ordering policy in our next simulation.

## 6.2. The Effect of Decreasing Review Period Length

Next, we investigate the impact of the review period on the priority-ordering policy performance for a fixed  $n$  (i.e., a nonasymptotic/high-volume setting). Whereas we characterized an asymptotically optimal rate for the review period as a function of  $n$  in Section 5.1, we show here that there is an optimal review period length in a finite system that balances the trade-off between matching quickly to avoid reneging and waiting to gather enough agents to make better matches. We numerically investigate this and demonstrate that both the network

**Figure 7.** (Color online) Uniform Distributed Patience Times with Mean  $1/3$  and Variance  $1/27$ 

**Figure 8.** Two Network Models



structure and patience time distributions play a critical role in the optimal review period length. We analyze two models with different network structures. To focus on the review period, we minimize the number of parameters in the systems we consider.

The two networks are depicted in Figure 8, (a) and (b), and are quite similar; Figure 8(b) is simply Figure 8(a) with supply node 2 removed. Thus, we describe the inputs for Figure 8(a) in detail and note that Figure 8(b) has all the same inputs excluding those for supply node 2. We compare these two models to illustrate how the network structure impacts the optimal review period length.

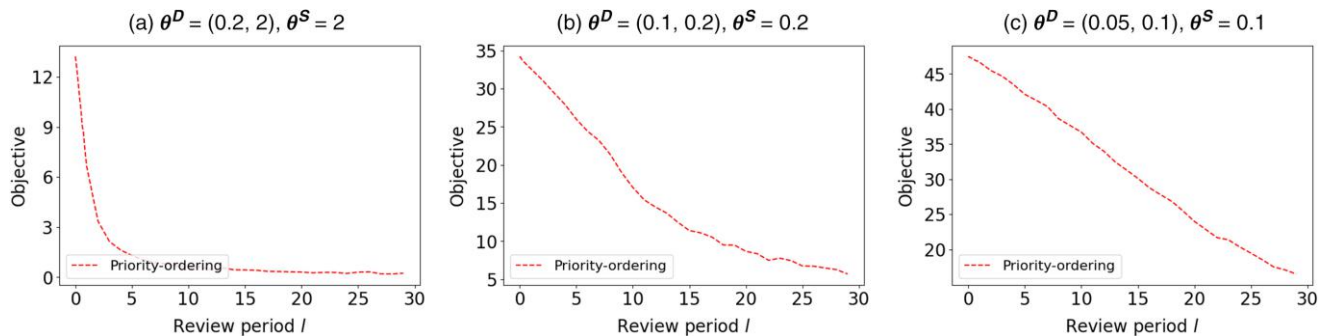
We now describe Figure 8(a), which consists of two demand and two supply nodes. We simulate three different cases with increasing patience times for each model, which we observe to impact the importance of the review period length. Each node has patience times that are uniformly distributed in  $[0, 2/\theta_j^D]$  and  $[0, 2/\theta_k^S]$ . Defining reneging rate vectors  $\theta^D = (\theta_1^D, \dots, \theta_j^D)$  and  $\theta^S = (\theta_1^S, \dots, \theta_k^S)$ , we consider the following three cases: (i)  $\theta^D = (0.2, 2)$  and  $\theta^S = (2, 0.2)$ , (ii)  $\theta^D = (0.1, 0.2)$  and  $\theta^S = (0.2, 0.1)$ , and (iii)  $\theta^D = (0.05, 0.1)$  and  $\theta^S = (0.1, 0.05)$ . For each of the three cases, we consider the same Poisson arrival rates  $\lambda^D = (1, 0.1)$  and  $\lambda^S = (0.1, 1)$  for the demand and supply nodes, respectively. We assume that the system is initially empty, consider  $v_1 = (0.1)$  and  $v_2 = (1, 0.1)$  as the value vectors for each

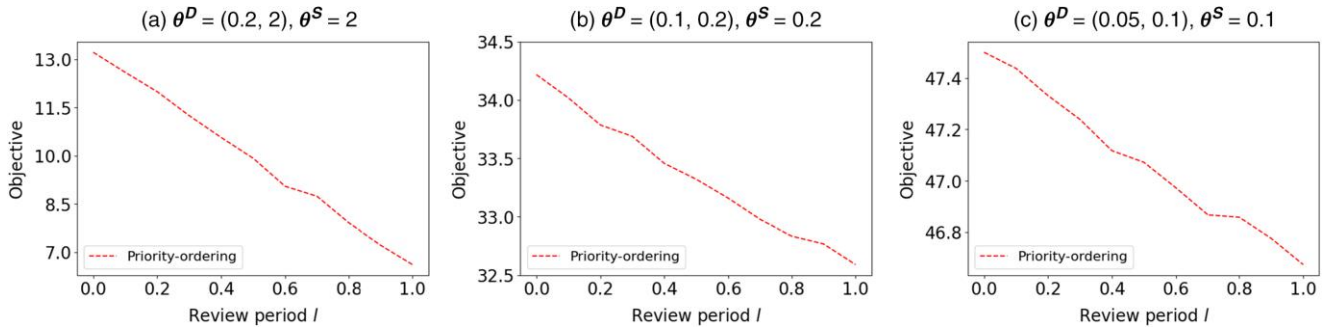
demand node, and assume no holding costs. We fix the time horizon  $t = 1,000$  in (1) and run 120 sample paths for each case.

**6.2.1. Demand Without Supply Alternatives.** We begin by simulating the priority ordering policy for the network in Figure 8(b), which has all the same inputs as Figure 8(a) with supply node 2 removed. For each case in this network, the MP solution simply makes all matches between demand node 2 and supply node 1, which we call high-value matches. Thus, the priority sets are  $\mathcal{P}_0 = \{(2, 1)\}$  and  $\mathcal{P}_1 = \{(1, 1)\}$ , and we call the second set low-value matches. Note that, for this network structure, if a worker at supply node 1 makes a low-value match with demand node 1, then we lose the ability to make a high-value match between that worker and a customer at demand node 2 in the future. Therefore, for this model, we refer to workers at supply node 1 and customers at demand node 2 as high-value agents and customers at demand node 1 as low-value agents.

We observe in this network that shorter review periods are better, regardless of the patience time distributions as illustrated in Figures 9 and 10 (the figures of this section show the objective value over the time horizon on the  $y$ -axis; we do not use the ratio compared with the upper bound because we are not considering the asymptotic regime). This is because customers at demand node 2 have no alternative match, so these

**Figure 9.** (Color online) Performance of the Priority-Ordering Policy for Model 2



**Figure 10.** (Color online) Performance of the Priority-Ordering Policy for Model 2

Note. Zoom into the x-axis between the values zero and one.

agents stay in the system until they renege or make a high value match with supply node 1 regardless of the review period length. Thus, with this network structure, the main focus of the policy should be on making matches quickly.

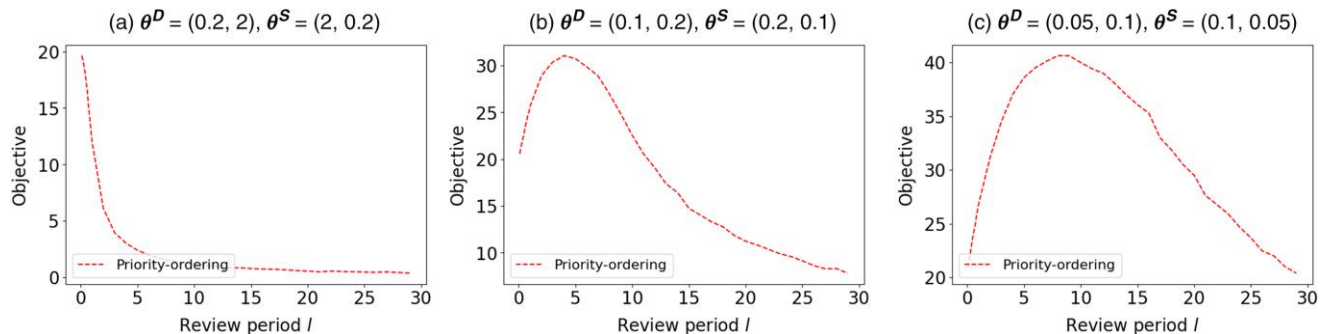
Following this line of thought, we may conjecture that for Figure 8(b) the review period length should be zero, meaning we make matches immediately when agents arrive if possible. We test this conjecture with an additional simulation for Figure 8(b) that focuses on review period lengths on a smaller scale,  $l \in [0, 1]$ . In particular, we simulate the priority policy for  $l \in \{0.1, 0.2, \dots, 1\}$  and also simulate a priority policy with  $l=0$ ; that is, a match is made (if possible) following the priority ordering each time an agent arrives to the system. These simulation results for Figure 8(b) are depicted in Figure 10, which demonstrate that the same intuition holds at this smaller scale as the review period converges to zero: shorter review periods are better because there is no benefit to waiting longer to gather more valuable matches in this system. This observation begs the question: are shorter review periods always better? We explore this question in the next set of simulations on Figure 8(a).

**6.2.2. Demand with Supply Alternatives.** We now simulate the priority ordering policy for Figure 8(a), which

has an additional supply node compared with Figure 8(b), which we see makes the length of the review period more important. For each case of this network, the MP solution simply makes all matches between demand node 2 and supply node 1, which we call high-value matches. Thus, the priority sets are  $\mathcal{P}_0 = \{(2, 1)\}$  and  $\mathcal{P}_1 = \{(1, 1), (2, 2)\}$ , and we call the second set low-value matches. With a similar reasoning to the one for Figure 8(b), for this model, we refer to customers at demand node 2 and workers at supply node 1 as high-value agents and customers at demand node 1 and workers at supply node 2 as low-value agents.

We analyze the impact of the length of the review period ( $l$ ) on the performance of the priority-ordering policy for each of the three cases and present the results in Figure 11. Next, we summarize the main insights from the simulations.

If the high-value agents' patience is too low, we do not allow valuable matches to arise, in which case, the smaller the review period, the better. In the first case, the short patience times (average of half a unit of time) and low arrival rates (average of one arrival every 10 time units) of the high-value agents make having both high-value demand and supply agents at the end of a review period unlikely. Therefore, in this case, waiting to gather valuable matches is unlikely to be fruitful, and

**Figure 11.** (Color online) Performance of the Priority-Ordering Policy for Model 1

the policy should instead focus on at least making low-value matches quickly before these agents renege. This is achieved with shorter review periods as low-value agents are abundant in the system because of their long patience times (average of five time units) and high arrival rates (average of one arrival every time unit). Thus, we see in the left panel of Figure 11 that the need to quickly match agents dominates in this case, and shorter review periods give better policy performance.<sup>7</sup>

If the high-value agents' patience is long enough to allow high-value matches to arise, then there is an optimal positive value for the review period length. In cases two and three, the high-value agents are patient enough to be present at the same time with higher likelihood. This is seen in the middle and right panels of Figure 11, in which, initially, increasing the review period allows gathering enough high-value agents to make more high-value matches, thus improving policy performance. This is different from Figure 8(b) because of the presence of supply node 2, which is able to steal high value customers away from demand node 2 and degrade performance if review periods are too short. Thus, in Figure 8(a), there is a true trade-off between making matches quickly and waiting to gather valuable matches because of the presence of low-value agents that can divert agents away from future high value matches.

The optimal value for the length of the review period depends on the high-value agents' mean patience times. Figure 11, (b) and (c), shows the bigger the high-value agents' mean patience times, the bigger the length of the optimal review period. Intuitively, when agents have small patience times, we need to decide matches faster not to lose the valuable matches. On the other hand, when high-value agents are willing to wait in the system, there is no rush to make matches and lose the opportunity to make high-value matches. The discrete review length should strike a balance between the arrival and reneging rates.

In summary, both the network and the patience time distribution critically impact the review period length. In general, the patience times need to be long enough relative to the arrival rates and review period length to accumulate enough valuable matches in a review period; however, in network structures that do not

allow stealing high-value agents along low-value arcs, shorter review periods may always be better. An interesting question for future research (that is beyond the scope of this paper) is to characterize the network structures for which an asymptotically optimal policy can make matches immediately when agents arrive as is the case numerically in Figure 10 and looks like the case in Figure 11(a) for appropriate conditions on the patience time distributions. A policy that makes matches immediately when agents arrive is a fully greedy policy.

## 7. A Matching Policy for a Model Without Holding Costs

The matching policies studied until now use the knowledge of the arrival rate parameters and the patience time distributions. This is because both policies use the information provided by the MP (7), which naturally depends on the invariant queue lengths. The latter uses the information of the arrival rate parameters and the patience time distributions.

However, there is an insensitivity property on the fraction of reneging agents. Table 1 shows the fraction of reneging agents obtained by the simulations and the corresponding values calculated by the fluid model equations (denoted by  $r_S^* := \lambda^D - \min(\lambda^D, \lambda^S)$  and  $r_D^* := \lambda^S - \min(\lambda^D, \lambda^S)$ ) in a network with one demand and one supply node that makes every possible match. We observe that, for exponential and gamma patience times, the fraction of reneging agents remains almost the same, which suggests an insensitivity property similar to ones that appear in Atar et al. (2022) and Puha and Ward (2019).

In this section, we sacrifice the generality of the objective function (1) to take advantage of the insensitivity property observed in Table 1. If we set  $c_j^D = c_k^S = 0$  for each  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$ , then (1) becomes the cumulative matching value of the platform, and hence, it is not affected by the number of customers/workers that renege. In other words, it is expected that the cumulative matching value is not affected by the patience time distribution because of the insensitivity property. This allows us to propose a discrete review matching policy that does not use the arrival rate and patience time distribution information.

**Table 1.** The Fraction of Reneging Population for Exponential (Right) and Gamma (Left, Rate Parameter = 0.7) Distributions with Unit Means

| $\lambda^S$ | $R^D(T)/A^D(T)$ | $r_D^*$ | $R^S(T)/A^S(T)$ | $r_S^*$ | $R^D(T)/A^D(T)$ | $r_D^*$ | $R^S(T)/A^S(T)$ | $r_S^*$ |
|-------------|-----------------|---------|-----------------|---------|-----------------|---------|-----------------|---------|
| 0.5         | 0.4942          | 0.5     | 1.4 e-5         | 0       | 0.4847          | 0.5     | 0               | 0       |
| 0.9         | 0.1193          | 0.1     | 0.0022          | 0       | 0.0959          | 0.1     | 0.0221          | 0       |
| 1.0         | 0.0586          | 0       | 0.0585          | 0       | 0.0130          | 0       | 0.0581          | 0       |
| 1.2         | 0.0066          | 0       | 0.173           | 0.2     | 0               | 0       | 0.1698          | 0.2     |
| 1.5         | 4.3 e-5         | 0       | 0.3314          | 0.5     | 0               | 0       | 0.3328          | 0.5     |

Note. The parameters are  $T = 100$ ,  $\lambda^D = 1$ , and  $n = 100$ .



In the case without holding costs, the MP takes the following form:

$$\begin{aligned} \max \quad & \sum_{(j,k) \in \mathcal{E}} v_{jk} m_{jk} \\ \text{s.t.} \quad & \sum_{j \in \mathcal{D}_k} m_{jk} \leq \lambda_k^S, k \in \mathbb{K}, \\ & \sum_{k \in \mathcal{S}_j} m_{jk} \leq \lambda_j^D, j \in \mathbb{J}, \\ & m_{jk} \geq 0, (j,k) \in \mathcal{E}. \end{aligned} \quad (35)$$

The number of matches made at each discrete review time point can be decided by solving an optimization problem with an objective to maximize the matching value and constraints that respect the amount of demand and supply available. For any  $i \geq 1$ , let  $(\mathcal{M}_{ijk}^{LP,*} : (j,k) \in \mathcal{E})$  be given by an optimal solution to the following optimization problem:

$$\begin{aligned} \max \quad & \sum_{(j,k) \in \mathcal{E}} v_{jk} \mathcal{M}_{ijk}^{LP} \\ \text{s.t.} \quad & \sum_{j \in \mathcal{D}_k} \mathcal{M}_{ijk}^{LP} \leq Q_k^S(il-), k \in \mathbb{K}, \\ & \sum_{k \in \mathcal{S}_j} \mathcal{M}_{ijk}^{LP} \leq Q_j^D(il-), j \in \mathbb{J}, \\ & \mathcal{M}_{ijk}^{LP} \in \mathbb{Z}_+, \text{ for all } (j,k) \in \mathcal{E}. \end{aligned} \quad (36)$$

The quantity  $\mathcal{M}_{ijk}^{LP,*}$  is the number of matches between type  $j$  demand and type  $k$  supply, and the cumulative number of matches for  $(j,k) \in \mathcal{E}$  and  $t \geq 0$  is given by

$$M_{jk}^{LP}(t) = \sum_{i=1}^{\lfloor t/l \rfloor} \mathcal{M}_{ijk}^{LP,*}, \quad (37)$$

where we refer to the preceding policy as the LP-based matching policy. It is straightforward to see that the LP-based matching policy satisfies the admissibility conditions given at the end of Section 2.

The main difference between (36) and (35) is that the right-hand side of the constraints in (35) are replaced by the queue lengths  $Q_j^D(il-) \in \mathbb{N}$  and  $Q_k^S(il-) \in \mathbb{N}$ . The proposed policy is myopic in the sense that the matching decisions made at each review time point are optimal given the available demand and supply but disregard the impact of future arrivals. The hope is that, when discrete review points are well-placed, the aforementioned myopicity will not have too much negative impact, and the resulting total value of matches made can be close to the optimal value to (35).

The implementation of the LP-based matching policy does not require the knowledge of the arrival rates  $\lambda^D, \lambda^S$  and the patience time distributions, but it requires an optimal solution to (36) at each review

point. From Lemma 3, (36) can effectively be solved as an LP on any sample path because the queue lengths are always integer valued.

**Lemma 3.** *The constraint matrix of (36) is totally unimodular, which implies that an optimal extreme point solution is integer-valued if the right-hand side constraints of (36) are integer-valued.*

Next, we consider the asymptotic behavior of the LP-based policy in the same asymptotic regime as in Section 5. The number of matches in the  $n$ th system at a discrete review period is given by an optimal solution to (36), replacing the right-hand sides of the constraints by  $Q_j^{D,n}(il^n-)$  and  $Q_k^{S,n}(il^n-)$ , respectively. The next theorem states that the matching policy is asymptotically optimal. Note that, in contrast to Theorems 3–5, the requirement that hazard rates be bounded is not needed. This is because, when holding costs are zero, there is not a need to analyze the queue-length behavior.

**Theorem 6.** *Assume  $c_j^D = c_k^S = 0$  for  $j \in \mathbb{J}$  and  $k \in \mathbb{K}$  and let  $\mathbf{m}^*$  be an optimal solution to (7).*

i. *If  $\{\mathbf{M}^n\}_{n \in \mathbb{N}}$  is a sequence of matching processes such that, as  $n \rightarrow \infty$ ,*

$$\sup_{0 \leq t \leq T} \left| \frac{M_{jk}^n(t)}{n} - m_{jk}^* \right| \rightarrow 0, \quad (38)$$

*in probability for some  $\mathbf{m} \in \mathbb{M}$  and for all  $T \geq 0$ , then*

$$\lim_{t \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{1}{nt} V_{\mathbf{M}^n}(t) = \sum_{(j,k) \in \mathcal{E}} v_{jk} m_{jk} \leq \sum_{(j,k) \in \mathcal{E}} v_{jk} m_{jk}^*. \quad (39)$$

ii. *The LP-based matching policy given by (36) and (37) in the  $n$ th system is asymptotically optimal in the sense that it satisfies*

$$\lim_{t \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{1}{nt} \frac{V_{\mathbf{M}^{LP,n}}(t)}{\sum_{(j,k) \in \mathcal{E}} v_{jk} m_{jk}^*} = 1,$$

*in probability.*

We prove Theorem 6 using the same three basic steps as Theorem 4. However, establishing a lower bound on the matches made during a review period (step ii) requires more effort because we cannot compare directly to the MP optimal solution. We overcome this challenge by leveraging a monotonicity property of the MP (35).

Having shown that the LP-based policy is asymptotically optimal, a question that arises is whether the matches made under this policy also achieve the optimal matching rates. The answer is nuanced in the case when there are multiple optimal solutions to the MP because the limit of  $\frac{M_{jk}^{LP,n}(t)}{n}$  may oscillate between them. However, we are able to show that the matching rates must approach the set of optimal MP solutions asymptotically. To this end, for a real vector  $x$  and a set  $A$  in Euclidean space, denote the distance between them by  $d(x, A)$ ; for example, one could consider  $d(x, A) := \inf_{z \in A} \|x - z\|_2$ .

**Proposition 4.** Let  $S$  be the set of all optimal solutions of (35). In other words, we have that  $S := \{m^* : m^* \text{ is an optimal solution to (35)}\}$ . Then, for each  $t > 0$  as  $n \rightarrow \infty$ ,

$$d\left(\frac{M^{LP,n}(t)}{nt}, S\right) \rightarrow 0,$$

in probability.

A consequence of the last theorem is that an analogous result to Theorem 4 holds for the LP-based policy in the special case that (35) has a unique optimal solution.

**Corollary 4.** Assume (35) has a unique optimal solution  $m^*$ . We have that, for each  $t \geq 0$  as  $n \rightarrow \infty$ ,

$$\frac{M_{jk}^{LP,n}(t)}{n} \rightarrow m_{jk}^* t,$$

in probability for all  $(j, k) \in \mathcal{E}$ .

## 8. Conclusion

In this paper, we propose and analyze a model that takes into account three main features of service platforms: (i) demand and supply heterogeneity, (ii) the random unfolding of arrivals over time, and (iii) the non-Markovian impatience of demand and supply. These features result in a trade-off between making a less good match quickly and waiting for a better match. The model is too complicated to solve for an optimal matching policy, and so we develop an approximating fluid model that is accurate at high volume (when demand and supply arrival rates are large). We use the invariant states of the fluid model to define a matching optimization problem, whose solution gives asymptotically optimal matching rates that depend on agent patience time distributions. We propose a discrete review policy to track those asymptotically optimal matching rates and further establish conditions under which a static priority ordering policy also results in asymptotically optimal matching rates. Finally, we observe that, when holding costs are zero, there is an insensitivity property that allows us to propose an LP-based matching policy that also achieves asymptotically optimal matching rates but does not depend on demand and supply patience time distributions.

## Acknowledgments

The authors thank Yuan Zhong for helpful discussions related to Lyapunov functions.

## Endnotes

<sup>1</sup> We henceforth use the term “agent” when collectively referring to customers and workers together.

<sup>2</sup> The proof that the solution to the MP motivates an asymptotically optimal matching policy in a high-volume regime (specifically, Theorems 4 and 5 in Section 5) requires the additional assumption that the hazard rates associated with the patience time distribution functions are bounded. However, our simulation results in Section 6 suggest that our proposed priority-ordering policies perform well

even when hazard rates are unbounded (as is true for the uniform distribution). See also Remark 3.

<sup>3</sup> For a precise mathematical specification of a matching policy and the admissible policy class, see definitions 1 and 2 in Aveklouris et al. (2023). A more rigorous mathematical specification is not possible in this paper because we do not fully specify the system dynamics.

<sup>4</sup> Note that  $\lim_{t \rightarrow \infty}(\bar{\eta}^D(t), \bar{\eta}^S(t)) = (\eta^{D,*}, \eta^{S,*})$  denotes the weak convergence of the measures; that is, for each  $j \in \mathbb{J}$  and all bounded  $f_j \in C([0, H_j^D])$ ,  $\lim_{t \rightarrow \infty} \langle f_j, \bar{\eta}_j^D(t) \rangle = \langle f_j, \eta_j^{D,*} \rangle$  and for each  $k \in \mathbb{K}$  and all bounded  $f_k \in C([0, H_k^S])$ ,  $\lim_{t \rightarrow \infty} \langle f_k, \bar{\eta}_k^S(t) \rangle = \langle f_k, \eta_k^{S,*} \rangle$ .

<sup>5</sup> The minimal assumption necessary should be finite  $(2 + \epsilon)$  moments, and then, the review period length should be defined in terms of  $\epsilon$ . We chose  $\epsilon = 3$  and avoided having  $\epsilon$  appear in the definition of  $l^n$ .

<sup>6</sup> Because arrival rates are of order  $n$ , the state descriptor should be divided by  $n$ .

<sup>7</sup> Note that, as the review period approaches zero, the objective corresponds to the fluid solution value when no high-value matches take place and the time horizon is  $t = 1,000$ .

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